```
// ===== FILE: Assets/Scripts/Game/Game.cs =====
using System;
using System.Collections;
using System.IO;
using Liangshan.Sim;
using UnityEngine;
namespace Liangshan
{
    /// <summary>
    /// 入口：空场景启动后自动挂上。负责镜头、主界面 <-> 战斗切换。
    /// 命令行带 -shots &lt;目录&gt; 时进入自检模式：机器人自动打一局，在关键节点截图后退出。
    /// </summary>
    public sealed class Game : MonoBehaviour
    {
        public static Game I { get; private set; }
        /// <summary>游戏自己的镜头。编辑器里打开的场景可能自带 Main Camera，不能用 Camera.main。</summary>
        public static Camera Cam { get; private set; }
        Camera _cam;
        RectTransform _floatLayer;
        BattleView _battle;
        HomeScreen _home;
        Vector2Int _screen;
        string _shots;
        /// <summary>只截局外界面（调界面布局时用，几十秒跑完）。</summary>
        bool _metaOnly;
        int _autoSkill = -1;
        [RuntimeInitializeOnLoadMethod(RuntimeInitializeLoadType.AfterSceneLoad)]
        static void Boot()
        {
            if (I != null) return;
            var go = new GameObject("Game");
            DontDestroyOnLoad(go);
            I = go.AddComponent<Game>();
        }
        void Awake()
        {
            Application.targetFrameRate = 60;
            Input.multiTouchEnabled = false;
            ParseArgs();
            var camGo = new GameObject("Camera");
            camGo.transform.SetParent(transform, false);
            _cam = camGo.AddComponent<Camera>();
            camGo.tag = "MainCamera";
            _cam.orthographic = true;
            _cam.clearFlags = CameraClearFlags.SolidColor;
            _cam.backgroundColor = Pal.Hex("1C2128");
            _cam.nearClipPlane = -50;
            _cam.farClipPlane = 50;
            _cam.depth = 100;
            camGo.AddComponent<AudioListener>();
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        Cam = _cam;
        MuteForeignCameras();
        FitCamera();
        Ui.Init(transform);
        Tween.Init(gameObject);
        Sfx.Init(gameObject);
        _floatLayer = Ui.Rect(Ui.Root, "FloatLayer");
        _floatLayer.anchorMin = Vector2.zero;
        _floatLayer.anchorMax = Vector2.one;
        _floatLayer.offsetMin = _floatLayer.offsetMax = Vector2.zero;
        WorldFx.Create(transform, _floatLayer, _cam.transform);
        Profile.Load();
        Sfx.On = Profile.S.Sound == 1;
        Sfx.Music = Profile.S.Music == 1;
        if (_shots != null) StartCoroutine(SelfCheck());
        else if (Application.isBatchMode) ShowHome();
        else
        {
            CloudSync.Imported += OnCloudImported;
            var boot = BootCover();
            CloudSync.Startup(() =>
            {
                Destroy(boot.gameObject);
                ShowHome();
            });
        }
    }
    /// <summary>等云存档（最多几秒）时盖住画面：先拉云再进首页，免得本地改动和云端打架。</summary>
    RectTransform BootCover()
    {
        var layer = Ui.Rect(Ui.Root, "boot");
        layer.anchorMin = Vector2.zero;
        layer.anchorMax = Vector2.one;
        layer.offsetMin = layer.offsetMax = Vector2.zero;
        var bg = Ui.Box(layer, new Vector2(1080, 1920), Vector2.zero, Color.white, Art.Img("h...
        bg.type = UnityEngine.UI.Image.Type.Simple;
        float scale = Mathf.Max(Ui.W / 1080f, Ui.Height / 1920f);
        bg.rectTransform.sizeDelta = new Vector2(1080, 1920) * scale;
        Ui.Fill(layer, new Color(0, 0, 0, 0.35f), true);
        Ui.Label(layer, "读取存档…", 46, Color.white, new Vector2(0, -Ui.Height * 0.5f + 260), ne...
        return layer;
    }
    void OnDestroy() => CloudSync.Imported -= OnCloudImported;
    /// <summary>局中被云端覆盖不打断，回首页时自然用新档。</summary>
    void OnCloudImported()
    {
        Sfx.On = Profile.S.Sound == 1;
        Sfx.Music = Profile.S.Music == 1;
        if (_home) ShowHome(_home.Tab);
    }
```

```
    void ParseArgs()
    {
        var args = Environment.GetCommandLineArgs();
        for (int i = 0; i < args.Length - 1; i++)
        {
            if (args[i] == "-shots") _shots = args[i + 1];
            if (args[i] == "-safe-top" && float.TryParse(args[i + 1], out float st)) Ui.FakeS...
            if (args[i] == "-safe-bottom" && float.TryParse(args[i + 1], out float sb)) Ui.Fa...
        }
        _metaOnly = Array.IndexOf(args, "-quickmeta") >= 0;
        if (_shots != null)
        {
            Directory.CreateDirectory(_shots);
            PlayerPrefs.DeleteAll();
            Profile.Volatile = true;
            Ads.Fast = true;
        }
    }
    void MuteForeignCameras()
    {
        foreach (var c in FindObjectsOfType<Camera>())
            if (c != _cam) c.enabled = false;
        foreach (var l in FindObjectsOfType<AudioListener>())
            if (l.gameObject != _cam.gameObject) l.enabled = false;
    }
    /// <summary>界面根节点让出底部横条后，战场跟着抬：世界 y = -9.6 对齐到横条上沿，横条里照常画战场。</summar
    void FitCamera()
    {
        _screen = new Vector2Int(Screen.width, Screen.height);
        float lift = Ui.SafeBottomPixels;
        float area = Mathf.Max(1f, Screen.height - lift);
        float aspect = area / Mathf.Max(1f, Screen.width);
        float half = aspect > 16f / 9f ? 5.4f * aspect : 9.6f;
        float unitsPerPixel = 2f * half / area;
        _cam.orthographicSize = Screen.height * unitsPerPixel * 0.5f;
        var p = new Vector3(0, -9.6f - lift * unitsPerPixel + _cam.orthographicSize, -10);
        _cam.transform.localPosition = p;
        if (WorldFx.I) WorldFx.I.SetCamBase(p);
    }
    void Update()
    {
        if (_screen.x != Screen.width || _screen.y != Screen.height)
        {
            FitCamera();
            Ui.Fit();
            // 首页按 SafeTop 排版，屏幕一变（微信启动头几帧常见）就整页重搭。
            if (_home) ShowHome(_home.Tab);
        }
        if ((_tick -= Time.unscaledDeltaTime) <= 0)
        {
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            _tick = 1f;
            int st = Profile.S.Stamina, day = Profile.S.Day;
            Profile.Tick();
            if (st != Profile.S.Stamina || day != Profile.S.Day) Profile.Save();
        }
    }
    float _tick;
    // ---------- 流程 ----------
    void Clear()
    {
        MuteForeignCameras();
        if (_battle) Destroy(_battle.gameObject);
        _battle = null;
        if (_home) Destroy(_home.gameObject);
        _home = null;
        Time.timeScale = 1f;
    }
    public HomeScreen Home => _home;
    /// <summary>首通解锁的新好汉：回到主页后全屏揭晓一次。</summary>
    HeroId? _pendingReveal;
    public void ShowHome(int tab = 2)
    {
        Clear();
        _home = HomeScreen.Create(tab);
        Sfx.PlayMusic(Bgm.Menu);
        if (_pendingReveal.HasValue && _autoSkill < 0)
        {
            var h = _pendingReveal.Value;
            var home = _home;
            Tween.Run(home, 0.35f, k => { }, () => MetaFx.HeroReveal(home.Layer, h));
        }
        _pendingReveal = null;
    }
    public void StartLevel(int no) => Play(Defs.Level(Mathf.Min(no, Profile.S.Cleared + 1)));
    public void StartDaily() => Play(Defs.DailyStage(Profile.S.Cleared, Profile.Today));
    public void StartEscort() => Play(Defs.EscortStage(Profile.S.Cleared));
    /// <summary>出征检查体力和次数；不够时弹补给窗或提示，返回 false。</summary>
    public bool Play(LevelDef st)
    {
        var s = Profile.S;
        bool auto = _autoSkill >= 0;
        if (st.Mode == StageMode.Daily && s.DailyTries >= Meta.DailyTriesMax && !auto) { Ui.T...
        if (st.Mode == StageMode.Escort && s.EscortTries >= Meta.EscortTriesMax && !auto) { U...
        if (!auto && !Meta.Spend(s, Cur.Stamina, Meta.BattleStamina))
        {
            if (_home) MetaUi.StaminaShop(_home.Layer);
            else
            {
                Ui.Toast("体力不足", 200);
                ShowHome();
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            }
            return false;
        }
        if (st.Mode == StageMode.Daily) s.DailyTries++;
        if (st.Mode == StageMode.Escort) s.EscortTries++;
        if (st.Mode == StageMode.Main) s.Selected = st.No;
        Profile.Save();
        Launch(st);
        return true;
    }
    /// <summary>重开本关：已经扣过体力，不再扣。</summary>
    public void Restart() => Launch(_stage);
    LevelDef _stage;
    int _seed;
    bool _settled;
    Rewards _last;
    void Launch(LevelDef st)
    {
        Clear();
        _stage = st;
        _settled = false;
        var go = new GameObject("Battle");
        go.transform.SetParent(transform, false);
        _battle = go.AddComponent<BattleView>();
        bool tutorial = _autoSkill < 0 && Profile.S.TutorialDone == 0 && st.Mode == StageMode...
        _seed = _autoSkill >= 0 ? 20261031 + _autoSkill : Environment.TickCount;
        _battle.Begin(_seed, st, Meta.BuildLoadout(Profile.S), tutorial);
        if (_autoSkill >= 0)
        {
            _battle.Auto = new Bot(_autoSkill);
            _battle.Speed = 2f;
        }
        else _battle.Speed = Profile.S.Speed15 == 1 && PlayerPrefs.GetInt("speed15", 0) == 1 ...
        _floatLayer.SetAsFirstSibling();
        Sfx.PlayMusic(st.Mode == StageMode.Main && st.IsChapterBoss ? Bgm.Boss : Bgm.Battle);
    }
    Rewards Compute(Battle b, bool win) =>
        Meta.BattleRewards(Profile.S, b.Stage, win, b.Wave, b.WallMax > 0 ? b.Wall / b.WallMa...
    public Rewards Preview(Battle b, bool win) => Compute(b, win);
    /// <summary>结算发奖：每局只发一次（失败后复活再赢，按赢算）。</summary>
    public Rewards Settle(Battle b, bool win)
    {
        if (_settled) return _last;
        _settled = true;
        var s = Profile.S;
        var r = Compute(b, win);
        Meta.Give(s, r);
        Meta.FinishMain(s, b.Stage, win, r);
        Meta.MarkSeen(s, b.Seen);
        Profile.Save();
```

```
        if (r.Unlock.HasValue && r.First) _pendingReveal = r.Unlock;
        _last = r;
        return r;
    }
    public void GiveAgain(Rewards r)
    {
        foreach (var it in r.Items) Meta.Give(Profile.S, it);
        Profile.Save();
    }
    // ---------- 自检截图 ----------
    bool _autoDone, _autoWon;
    public void OnAutoDone(bool won)
    {
        _autoDone = true;
        _autoWon = won;
    }
    void Shot(string name)
    {
        var path = Path.Combine(_shots, name + ".png");
        ScreenCapture.CaptureScreenshot(path);
        Debug.Log("[自检] 截图 " + path);
    }
    /// <summary>
    /// 走路自检：1 倍速，用一台特写相机跟住一个正在走的怪，每 3 帧拍一张共 16 张，
    /// 拼起来能直接看出两腿是否交替迈步、身体是不是在弹。
    /// </summary>
    IEnumerator WalkBurst(EnemyKind kind, string prefix)
    {
        Enemy target = null;
        foreach (var e in _battle.B.Enemies)
            if (e.Def.Kind == kind && !e.Attacking && e.Stun <= 0 && (target == null || e.Pos...
        if (target == null) yield break;
        float keep = _battle.Speed;
        _battle.Speed = 1f;
        var cam = new GameObject("walkCam").AddComponent<Camera>();
        cam.CopyFrom(_cam);
        cam.depth = _cam.depth + 1;
        cam.orthographicSize = target.Def.Size * 1.6f;
        for (int i = 0; i < 16; i++)
        {
            for (int f = 0; f < 3; f++)
            {
                var p = target.Pos.ToV();
                cam.transform.position = new Vector3(p.x, p.y + target.Def.Size * 0.5f, -10);
                yield return null;
            }
            Shot(prefix + i.ToString("00"));
        }
        yield return null;
        Destroy(cam.gameObject);
```

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        _battle.Speed = keep;
    }
    IEnumerator WaitReal(float s)
    {
        float end = Time.realtimeSinceStartup + s;
        while (Time.realtimeSinceStartup < end) yield return null;
    }
    /// <summary>按真人路径操作一遍：点召唤、按引导拖拽合成、点强化选卡。指针走 BattleView 的输入代码。</summa
    IEnumerator ManualPlay()
    {
        var v = _battle;
        v.B.Coins += 600;
        v.Hud.PressSummon();
        yield return WaitReal(0.5f);
        for (int i = 0; i < 8 && !v.FindMergePair(out _, out _); i++)
        {
            v.Hud.PressSummon();
            yield return WaitReal(0.25f);
        }
        yield return WaitReal(1.0f);
        Shot("02b_merge_guide");
        yield return WaitReal(0.3f);
        int a = -1, b = -1;
        for (int i = 0; i < v.B.Slots.Count && a < 0; i++)
            for (int j = 0; j < v.B.Slots.Count; j++)
                if (v.B.CanMerge(i, j)) { a = i; b = j; break; }
        if (a < 0)
        {
            Debug.LogError("[自检] 没凑出可合成的一对");
            yield break;
        }
        var from = v.SlotWorld(a);
        var to = v.SlotWorld(b);
        var ptr = new BattleView.Pointer { Down = true, Held = true, World = from };
        v.FakePointer = () => ptr;
        yield return null;
        ptr.Down = false;
        float t0 = Time.realtimeSinceStartup;
        bool shot = false;
        while (Time.realtimeSinceStartup - t0 < 0.7f)
        {
            float k = (Time.realtimeSinceStartup - t0) / 0.7f;
            ptr.World = Vector2.Lerp(from, to, k) + new Vector2(0, Mathf.Sin(k * Mathf.PI) * ...
            if (!shot && k > 0.85f) { shot = true; Shot("02c_dragging"); }
            yield return null;
        }
        ptr.World = to;
        yield return null;
        int mergesBefore = v.MergesDone;
        ptr.Held = false;
```

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        ptr.Up = true;
        yield return null;
        v.FakePointer = null;
        yield return WaitReal(0.35f);
        Shot("02d_merged");
        Debug.Log($"[自检] 拖拽合成 {(v.MergesDone > mergesBefore || v.B.Slots[b].Unit?.Star > 1 ? ...
        yield return WaitReal(0.6f);
        Shot("02d2_book_guide");
        yield return WaitReal(0.3f);
        v.Hud.PressEnhance();
        yield return WaitReal(1.0f);
        Shot("02e_enhance_cards");
        v.Hud.ChooseCard(0);
        yield return WaitReal(1.2f);
        Shot("02f_after_pick");
        yield return WaitReal(0.3f);
        Slot info = null;
        foreach (var sl in v.B.Slots)
            if (sl.Unit != null && (info == null || sl.Unit.Star > info.Unit.Star)) info = sl;
        if (info != null)
        {
            v.Hud.SlotInfo(info);
            yield return WaitReal(0.5f);
            Shot("02i_unit_info");
            yield return WaitReal(0.2f);
            v.Hud.SlotInfo(info, 1);
            yield return WaitReal(0.3f);
            Shot("02j_unit_ult");
            yield return WaitReal(0.2f);
            v.Hud.CloseInfo();
        }
        v.Hud.LordInfo();
        yield return WaitReal(0.45f);
        Shot("02k_lord");
        yield return WaitReal(0.35f);
        v.Hud.CloseInfo();
        var who = v.B.Lineup[0];
        PerkDef plain = null, extra = null;
        foreach (var p in Defs.Perks)
        {
            if (p.Hero != who) continue;
            if (!p.Ult && !p.Star && plain == null) plain = p;
            else if (p.Ult) extra = p;
            else if (p.Star && extra == null) extra = p;
        }
        if (plain != null) v.B.Perks[plain.Id] = 1;
        if (extra != null) v.B.Perks[extra.Id] = 1;
        yield return null;
        v.Hud.OpenPerks(who);
        yield return WaitReal(0.45f);
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        Shot("02l_perks");
        yield return WaitReal(0.35f);
        v.Hud.ClosePerk();
        yield return WaitReal(0.3f);
        v.Hud.OpenPanel("stats");
        yield return WaitReal(1.0f);
        Shot("02g_stats");
        v.Hud.ClosePanel();
        yield return WaitReal(0.3f);
        v.Hud.OpenPanel("pause");
        yield return WaitReal(0.6f);
        Shot("02h_pause");
        v.Hud.ClosePanel();
        yield return WaitReal(0.3f);
    }
    void CloseModals()
    {
        if (_home == null) return;
        for (int i = _home.Layer.childCount - 1; i >= 0; i--) DestroyImmediate(_home.Layer.Ge...
    }
    /// <summary>截图在帧末才落盘：截完等一下再动界面，免得拍到下一步的半成品。</summary>
    IEnumerator Snap(string name)
    {
        Shot(name);
        yield return WaitReal(0.15f);
    }
    IEnumerator ModalShot(Action open, string name)
    {
        CloseModals();
        open();
        yield return WaitReal(0.9f);
        yield return Snap(name);
    }
    /// <summary>局外各界面截图：先把存档推到“玩了一阵子”的样子。</summary>
    IEnumerator MetaShots()
    {
        _autoSkill = -1;
        var s = Meta.ExpectedSave(12);
        s.Cleared = 11;
        s.Selected = 12;
        for (int i = 0; i < 11; i++)
        {
            s.Stars[i] = i % 4 == 0 ? 2 : 3;
            s.LevelChest[i] = i % 4 == 0 ? 3 : 7;
        }
        s.Pages = 140;
        s.Tomes = 4;
        s.Silk = 2600;
        s.Meteor = 5;
        s.GearBoxes = 3;
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            s.PageLv = 4;
            s.TomeNodes = 1;
            s.FlagLv = 3;
            s.FlagAffix[0] = 0 * 4 + 2 + 1;
            s.FlagAffix[1] = 7 * 4 + 1 + 1;
            s.FlagLock = 1;
            s.EquipQ[0] = 2;
            s.EquipQ[3] = 1;
            s.Meals = 1;
            s.HeroLv[(int)HeroId.ZhangQing] = 3;
            s.HeroLv[(int)HeroId.WuYong] = 2;
            foreach (var h in new[] { HeroId.HuSanniang, HeroId.QinMing, HeroId.AnDaoquan, HeroId...
                s.HeroLv[(int)h] = Mathf.Max(1, s.HeroLv[(int)h]);
            s.Shards[(int)Meta.Starters[0]] = 30;
            s.Shards[(int)Meta.Starters[2]] = 5;
            s.Tokens = 14;
            s.Ingots = 860;
            s.Copper = 26000;
            s.Iron = 120;
            s.Stamina = 37;
            s.StaminaAt = Profile.Now - 100;
            s.IdleAt = Profile.Now - 3 * 3600 - 1200;
            s.Seen[(int)EnemyKind.Soldier] = 2;
            s.Seen[(int)EnemyKind.Shield] = 1;
            s.Seen[(int)EnemyKind.Cavalry] = 1;
            s.Seen[(int)EnemyKind.Porter] = 2;
            s.Seen[(int)EnemyKind.Hawk] = 1;
            s.Task[0] = 3;
            s.Task[1] = 2;
            s.Task[4] = 1;
            s.SignDays = 2;
            s.SignDay = Profile.Today - 1;
            s.Day = Profile.Today;
            Profile.Use(s);
            ShowHome(HomeScreen.TabBattle);
            yield return WaitReal(1.5f);
            yield return Snap("13_home_ch2");
            yield return ModalShot(() => MetaUi.SignIn(_home.Layer), "14_sign");
            yield return ModalShot(() => MetaUi.Tasks(_home.Layer), "15_tasks");
            yield return ModalShot(() => MetaUi.Idle(_home.Layer), "16_idle");
            yield return ModalShot(() => MetaUi.Codex(_home.Layer), "17_codex");
            yield return ModalShot(() => MetaUi.Daily(_home.Layer), "18_daily");
            yield return ModalShot(() => MetaUi.Escort(_home.Layer), "18b_escort");
            yield return ModalShot(() => MetaUi.StaminaShop(_home.Layer), "18c_stamina");
            yield return ModalShot(() => MetaUi.Chest(_home.Layer, 1, 2), "18d_chest");
            var chest2 = _home.transform.Find("page/body/chests/chest2");
            if (chest2)
            {
                yield return ModalShot(() => MetaUi.LevelChestCard(_home.Layer, 15, 2, chest2), "...
                yield return ModalShot(() =>
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        {
            MetaUi.LevelChestCard(_home.Layer, 15, 2, chest2);
            var tile = _home.Layer.Find("pop_chest/card/item");
            MetaUi.ItemTip(_home.Layer, new Item(Cur.Tome, 3), tile ? tile : chest2);
        }, "18f_item_tip");
    }
    yield return ModalShot(() => MetaUi.LevelList(_home.Layer), "18g_levels");
    yield return ModalShot(() => MetaUi.Meals(_home.Layer), "18h_meals");
    yield return ModalShot(() => MetaUi.ChapterChests(_home.Layer, 2), "18i_chapter_chest...
    CloseModals();
    s.Selected = 11;
    Profile.Save();
    yield return WaitReal(0.8f);
    yield return Snap("13b_home_sweep");
    s.Selected = 12;
    Profile.Save();
    yield return WaitReal(0.5f);
    _home.SetTab(HomeScreen.TabHero);
    yield return WaitReal(1.0f);
    yield return Snap("19_heroes");
    yield return ModalShot(() => MetaUi.HeroDetail(_home.Layer, HeroId.HuaRong), "20_hero...
    yield return ModalShot(() => MetaUi.HeroDetail(_home.Layer, HeroId.LinChong), "20b_he...
    yield return ModalShot(() => MetaUi.HeroDetail(_home.Layer, HeroId.HuSanniang), "20c_...
    yield return ModalShot(() => MetaUi.HeroDetail(_home.Layer, Meta.Starters[0]), "20d_h...
    yield return ModalShot(() => MetaUi.GetPaths(_home.Layer, Meta.Starters[1]), "20e_get...
    CloseModals();
    _home.BeginSwap(HeroId.ZhangQing);
    yield return WaitReal(0.8f);
    yield return Snap("20f_swap");
    _home.SetTab(HomeScreen.TabBattle);
    _home.SetTab(HomeScreen.TabHero);
    var lh = Meta.Starters[0];
    MetaFx.LevelUp(_home.Layer, new MetaFx.LevelUpInfo { Hero = lh, NewLv = 5, OldPower =...
    yield return WaitReal(1.4f);
    yield return Snap("20g_levelup");
    yield return WaitReal(2.2f);
    yield return Snap("20h_perma");
    CloseModals();
    MetaFx.HeroReveal(_home.Layer, HeroId.HuaRong);
    yield return WaitReal(1.6f);
    yield return Snap("20i_reveal");
    CloseModals();
    _home.SetTab(HomeScreen.TabForge);
    yield return WaitReal(1.0f);
    yield return Snap("21_forge");
    yield return ModalShot(() => MetaUi.ForgeSlot(_home.Layer, 0), "22_forge_slot");
    yield return ModalShot(() => MetaUi.TalentTree(_home.Layer), "22b_talent");
    yield return ModalShot(() => MetaUi.Tome(_home.Layer), "22c_tome");
    yield return ModalShot(() => MetaUi.Flag(_home.Layer), "22d_flag");
    yield return ModalShot(() => MetaUi.GearBox(_home.Layer), "22e_gearbox");
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        CloseModals();
        _home.SetTab(HomeScreen.TabFort);
        yield return WaitReal(1.0f);
        yield return Snap("23_fort");
        _home.SetTab(HomeScreen.TabShop);
        yield return WaitReal(1.0f);
        yield return Snap("24_shop");
        _home.DoRecruit(10, false);
        yield return WaitReal(2.2f);
        yield return Snap("24b_recruit10");
        CloseModals();
    }
    /// <summary>后期关卡：按标准养成让机器人打一局，看新敌人、新首领和章节战场。</summary>
    IEnumerator AutoBattle(int level, string prefix)
    {
        var s = Meta.ExpectedSave(level);
        s.HeroLv[(int)HeroId.ZhangQing] = Mathf.Max(1, s.HeroLv[(int)HeroId.ZhangQing]);
        s.Lineup = level >= 40 ? new[] { 2, 6, 5, 7 } : new[] { 0, 2, 4, 1 };
        for (int h = 0; h < Defs.HeroCount; h++) if (Array.IndexOf(s.Lineup, h) >= 0) s.HeroL...
        Profile.Use(s);
        _autoSkill = 2;
        Play(Defs.Level(level));
        _battle.Speed = 3f;
        bool boss = false, elite = false;
        float last = 0;
        int n = 0;
        while (true)
        {
            var b = _battle.B;
            var hud = _battle.Hud;
            if (!elite && hud.ModalKind == null)
                foreach (var e in b.Enemies)
                    if (e.Def.Rank == EnemyRank.Elite && e.Pos.Y < 4f) { elite = true; Shot(p...
            if (!boss && b.Wave == b.Stage.BossWave && b.Time > (b.Stage.BossWave - 1) * Defs...
            if (b.Time - last > 40f && hud.ModalKind == null)
            {
                last = b.Time;
                Shot($"{prefix}_p{n++}");
            }
            if (b.Phase == Phase.Victory || b.Phase == Phase.Defeat)
            {
                while (hud.ModalKind != "victory" && hud.ModalKind != "defeat") yield return ...
                yield return WaitReal(1.4f);
                Shot(prefix + "_" + (b.Phase == Phase.Victory ? "win" : "lose"));
                Debug.Log($"[自检] 第{level}关 {b.Phase}，墙 {b.Wall:0}/{b.WallMax:0}，用时 {b.Time:0}...
                break;
            }
            yield return null;
        }
        _autoDone = false;
```

```
        yield return WaitReal(0.3f);
    }
    IEnumerator SelfCheck()
    {
        if (_metaOnly)
        {
            Sfx.On = false;
            Sfx.Music = false;
            _autoSkill = 0;
            Play(Defs.Level(1));
            yield return WaitReal(2f);
            yield return MetaShots();
            Application.Quit(0);
            yield break;
        }
        Sfx.On = false;
        Sfx.Music = false;
        // 模拟编辑器场景里自带的 Main Camera，确认不会抢走画面。
        var decoy = new GameObject("Main Camera") { tag = "MainCamera" }.AddComponent<Camera>();
        decoy.orthographic = true;
        decoy.orthographicSize = 5;
        decoy.gameObject.AddComponent<AudioListener>();
        ShowHome();
        yield return WaitReal(1.5f);
        Shot("01_home");
        yield return WaitReal(0.5f);
        // 新手引导截一张：真人模式开局。
        StartLevel(1);
        yield return WaitReal(2.5f);
        Shot("02_tutorial");
        yield return WaitReal(0.4f);
        yield return ManualPlay();
        _autoSkill = 2;
        Play(Defs.Level(1));
        bool cards = false, reels = false, boss = false, rage = false, mid = false, early = f...
        float lastPeriodic = 0;
        int periodic = 0, lordCasts = 0;
        _autoDone = false;
        while (!_autoDone)
        {
            var b = _battle.B;
            var hud = _battle.Hud;
            if (!early && b.Time > 6f)
            {
                early = true;
                Shot("03_early");
                yield return WalkBurst(EnemyKind.Soldier, "03w");
            }
            if (!walkBig && b.Wave >= 3 && hud.ModalKind == null)
            {
```

```
            foreach (var en in b.Enemies)
                if (en.Def.Kind == EnemyKind.Shield && !en.Attacking && en.Pos.Y < 6f) { ...
            if (walkBig) yield return WalkBurst(EnemyKind.Shield, "03s");
        }
        if (!gate && b.Wave == 2 && _battle.GateOpen > 0.95f && hud.ModalKind == null) { ...
        if (lordCasts < 2 && b.LordCasts > lordCasts && hud.ModalKind == null)
        {
            lordCasts = b.LordCasts;
            for (int i = 0; i < 3; i++)
            {
                yield return WaitReal(0.12f);
                Shot($"03c_lord{lordCasts}_{i}");
            }
        }
        if (!cards && hud.ModalKind == "cards")
        {
            cards = true;
            yield return WaitReal(0.9f);
            Shot("04_cards");
        }
        if (!reels && hud.ModalKind == "reels")
        {
            reels = true;
            yield return WaitReal(0.5f);
            Shot("05_reels_open");
            yield return WaitReal(2.8f);
            Shot("06_reels_result");
        }
        if (!mid && b.Time > 45f && hud.ModalKind == null)
        {
            mid = true;
            Shot("07_mid");
            for (int i = 0; i < 3; i++)
            {
                yield return WaitReal(0.12f);
                Shot("07_fx" + i);
            }
        }
        if (!boss && b.Wave == b.Stage.BossWave && b.Time > (b.Stage.BossWave - 1) * Defs...
        if (!rage && b.Wave == b.Stage.RageWave && b.Time > (b.Stage.RageWave - 1) * Defs...
        if (!late && b.Wave >= 10 && b.Time > 9 * Defs.WaveTime + 8f && hud.ModalKind == ...
        if (b.Time - lastPeriodic > 30f && hud.ModalKind == null)
        {
            lastPeriodic = b.Time;
            Shot($"p{periodic++:00}_t{Mathf.RoundToInt(b.Time)}");
        }
        if (b.Phase == Phase.Victory || b.Phase == Phase.Defeat)
        {
            while (hud.ModalKind != "victory" && hud.ModalKind != "defeat") yield return ...
            yield return WaitReal(1.4f);
```

```
            Shot(b.Phase == Phase.Victory ? "11_victory" : "11_defeat_skill2");
            Debug.Log($"[自检] 第一局结果 {b.Phase}，墙 {b.Wall:0}/{b.WallMax:0}，用时 {b.Time:0}s");
            yield return WaitReal(0.4f);
            break;
          }
          yield return null;
        }
        // 失败结算：只会召唤的机器人。
        _autoSkill = 0;
        Play(Defs.Level(1));
        _battle.Speed = 3f;
        while (true)
        {
          var b = _battle.B;
          if (b.Phase == Phase.Defeat || b.Phase == Phase.Victory)
          {
            while (_battle.Hud.ModalKind != "victory" && _battle.Hud.ModalKind != "defeat...
            yield return WaitReal(1.2f);
            Shot("12_" + (b.Phase == Phase.Defeat ? "defeat" : "victory_skill0"));
            break;
          }
          yield return null;
        }
        yield return WaitReal(0.5f);
        yield return MetaShots();
        yield return AutoBattle(35, "25_L35");
        yield return AutoBattle(50, "26_L50");
        Debug.Log("[自检] 完成");
        Application.Quit(0);
      }
    }
}
// ===== FILE: Assets/Scripts/Game/HomeScreen.cs =====
using System;
using System.Collections.Generic;
using Liangshan.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Liangshan
{
    /// <summary>主界面：顶栏资源、五个页签（商城/好汉/出征/寨主/山寨），弹窗都挂在 Layer 上。</summary>
    public sealed class HomeScreen : MonoBehaviour
    {
        public const int TabShop = 0, TabHero = 1, TabBattle = 2, TabForge = 3, TabFort = 4;
        RectTransform _root, _page;
        public RectTransform Layer { get; private set; }
        int _tab = TabBattle;
        /// <summary>好汉页职业筛选：-1 全部。</summary>
        int _roleFilter = -1;
        float _rosterScroll;
```

```
    bool _dirty;
    Text _staminaText, _staminaWait;
    float _tickT;
    public int Tab => _tab;
    static SaveData S => Profile.S;
    public static HomeScreen Create(int tab)
    {
        var rt = Ui.Rect(Ui.Root, "Home");
        Stretch(rt);
        var h = rt.gameObject.AddComponent<HomeScreen>();
        h._root = rt;
        h._tab = tab;
        h.Layer = Ui.Rect(rt, "layer");
        Stretch(h.Layer);
        h.Rebuild(true);
        Profile.Changed += h.MarkDirty;
        return h;
    }
    void OnDestroy() => Profile.Changed -= MarkDirty;
    void MarkDirty() => _dirty = true;
    static void Stretch(RectTransform rt)
    {
        rt.anchorMin = Vector2.zero;
        rt.anchorMax = Vector2.one;
        rt.offsetMin = rt.offsetMax = Vector2.zero;
    }
    float _slideDir;
    public void SetTab(int t)
    {
        if (t == _tab) return;
        _slideDir = t > _tab ? 1f : -1f;
        _tab = t;
        _swap = null;
        Sfx.Play(Snd.Tab, 0.8f);
        Rebuild(true);
    }
    /// <summary>任务“前往”：跳到能完成该任务的地方。</summary>
    public void Goto(int task)
    {
        int[] tabs = { TabBattle, TabBattle, TabHero, TabForge, TabShop, TabShop };
        SetTab(tabs[Mathf.Clamp(task, 0, tabs.Length - 1)]);
    }
    void LateUpdate()
    {
        if (_dirty)
        {
            _dirty = false;
            Rebuild(false);
        }
        if ((_tickT -= Time.unscaledDeltaTime) <= 0 && _staminaText)
```

```
        {
            _tickT = 0.5f;
            _staminaText.text = $"{S.Stamina}/{Meta.StaminaMax}";
            int w = Meta.StaminaWait(S, Profile.Now);
            _staminaWait.text = S.Stamina >= Meta.StaminaMax ? "" : $"{w / 60}:{w % 60:00}";
        }
    }
    void Rebuild(bool anim)
    {
        if (_page) DestroyImmediate(_page.gameObject);
        _page = Ui.Rect(_root, "page");
        Stretch(_page);
        _page.SetSiblingIndex(0);
        switch (_tab)
        {
            case TabShop: BuildShop(anim); break;
            case TabHero: BuildHeroes(anim); break;
            case TabForge: BuildForge(anim); break;
            case TabFort: BuildFort(anim); break;
            default: BuildBattle(anim); break;
        }
        // 页面内容收进 body，切页时只让 body 淡入滑入，顶栏和页签不闪
        var body = Ui.Rect(_page, "body");
        Stretch(body);
        while (_page.childCount > 1) _page.GetChild(0).SetParent(body, false);
        if (anim)
        {
            var cg = body.gameObject.AddComponent<CanvasGroup>();
            float dir = _slideDir;
            Tween.Run(body, 0.2f, k =>
            {
                cg.alpha = Mathf.Lerp(0.2f, 1f, k);
                body.anchoredPosition = new Vector2(dir * 60f * (1 - Tween.OutCubic(k)), 0);
            });
        }
        BuildTopBar();
        BuildTabs();
        Layer.SetAsLastSibling();
    }
    RectTransform Top(float x, float y, Vector2 size, string name) => Ui.Top(_page, x, y + Ui...
    RectTransform Bottom(float x, float y, Vector2 size, string name) => Ui.Bottom(_page, x, ...
    /// <summary>背景按屏幕铺满（cover），返回图片实际尺寸。</summary>
    Vector2 Cover(string art, Vector2 natural, Color tint, float centerY = 0.5f)
    {
        var bg = Ui.Box(_page, natural, Vector2.zero, tint, Art.Img(art), "bg");
        bg.type = Image.Type.Simple;
        float scale = Mathf.Max(Ui.W / natural.x, Ui.Height / natural.y);
        var size = natural * scale;
        bg.rectTransform.sizeDelta = size;
        bg.rectTransform.anchoredPosition = new Vector2(0, (centerY - 0.5f) * (size.y - Ui.He...
```

```
851         return size;
852     }
853     /// <summary>顶、底各压一层淡蓝渐变，只为托住顶栏和页签上的白字，不把画面压暗。</summary>
854     void Shade(float top, float bottom)
855     {
856         var tint = Pal.Navy;
857         if (top > 0)
858         {
859             var t = Ui.Box(_page, new Vector2(1080, 420), Vector2.zero, tint.A(top), Art.VGra...
860             t.rectTransform.anchorMin = new Vector2(0, 1);
861             t.rectTransform.anchorMax = new Vector2(1, 1);
862             t.rectTransform.sizeDelta = new Vector2(0, 420);
863             t.rectTransform.anchoredPosition = new Vector2(0, -210);
864             t.rectTransform.localScale = new Vector3(1, -1, 1);
865         }
866         if (bottom > 0)
867         {
868             var b = Ui.Box(_page, new Vector2(1080, 760), Vector2.zero, tint.A(bottom), Art.V...
869             b.rectTransform.anchorMin = new Vector2(0, 0);
870             b.rectTransform.anchorMax = new Vector2(1, 0);
871             b.rectTransform.sizeDelta = new Vector2(0, 760);
872             b.rectTransform.anchoredPosition = new Vector2(0, 380);
873         }
874     }
875     // ---------- 顶栏、页签 ----------
876     static readonly Color Glass = new Color(0.05f, 0.09f, 0.16f, 0.5f);
877     void BuildTopBar()
878     {
879         float y = Ui.SafeTop + 62;
880         // 头像：金框、红底等级章、名字木牌、经验条
881         var av = Ui.Top(_page, -470, y, new Vector2(116, 116), "avatar");
882         var frame = Ui.Icon(av, "ui/h_avatar", 116, Vector2.zero, "frame");
883         Ui.Icon(av, "prop_songjiang", 96, new Vector2(0, 2));
884         int lv = Meta.LordLevel(S, out float xp);
885         Ui.Icon(av, "ui/h_round", 46, new Vector2(-44, -42), "lvBadge");
886         Ui.Outlined(av, lv.ToString(), 24, Color.white, new Vector2(-44, -41), new Vector2(46...
887         Ui.Tap(frame, () => SetTab(TabForge), av);
888         var nm = Ui.Top(_page, -318, y - 18, new Vector2(176, 50), "name");
889         Ui.Skin(nm, "h_plaque", new Vector2(176, 50), Vector2.zero, -1f);
890         Ui.Outlined(nm, "宋江", 28, Pal.Hex("FFE08A"), new Vector2(0, 1), new Vector2(160, 46),...
891         var xb = Ui.Top(_page, -318, y + 22, new Vector2(170, 18), "xp");
892         Ui.Box(xb, new Vector2(164, 18), Vector2.zero, Pal.Hex("3A0E06").A(0.85f), Art.Pill);
893         var xf = Ui.Box(xb, new Vector2(Mathf.Max(16, 158 * xp), 12), new Vector2(-79, 0), Pa...
894         xf.rectTransform.pivot = new Vector2(0, 0.5f);
895         CurPill(-50, y - 14, "ic_ingot", Ui.Num(S.Ingots), () => SetTab(TabShop));
896         CurPill(184, y - 14, "ic_coins", Ui.Num(S.Copper), null);
897         var st = CurPill(414, y - 14, "ic_wine", $"{S.Stamina}/{Meta.StaminaMax}", () => Meta...
898         _staminaText = st.Find("text").GetComponent<Text>();
899         _staminaWait = Ui.Outlined(st, "", 20, Pal.Hex("C8F0B0"), new Vector2(8, -40), new Ve...
900         _tickT = 0;
```

```
    }
    static readonly Color Lacquer = Pal.Hex("5A1E0C");
    /// <summary>资源胶囊：奶白面、朱红描边，深棕字；左侧图标压边，右侧绿色加号。</summary>
    RectTransform CurPill(float x, float y, string icon, string text, Action plus)
    {
        var c = Ui.Top(_page, x, y, new Vector2(204, 56), "cur_" + icon);
        Ui.Skin(c, "h_capsule", new Vector2(188, 52), new Vector2(8, 0), -1f, null, "bg");
        Ui.Icon(c, icon, 64, new Vector2(-82, 2));
        Ui.Label(c, text, 27, Lacquer, new Vector2(plus != null ? 4 : 14, 1), new Vector2(130...
        if (plus != null)
        {
            var p = Ui.Skin(c, "btn_green", new Vector2(40, 40), new Vector2(84, 0), -1f);
            Ui.Outlined(p.transform, "+", 30, Color.white, new Vector2(0, 2), new Vector2(40,...
            Ui.Tap(p, plus, c);
        }
        return c;
    }
    public const float TabH = 156;
    void BuildTabs()
    {
        var bar = Ui.Rect(_page, "tabs");
        bar.anchorMin = new Vector2(0, 0);
        bar.anchorMax = new Vector2(1, 0);
        bar.sizeDelta = new Vector2(0, TabH);
        bar.anchoredPosition = new Vector2(0, TabH * 0.5f);
        // 横条底下一直铺到屏幕底边（横条区也是这块木板），上沿金边压在页签区顶上。
        var under = Ui.Box(bar, new Vector2(1080, TabH), Vector2.zero, Pal.Hex("6E1C10"), Art...
        Stretch(under.rectTransform);
        under.rectTransform.offsetMin = new Vector2(0, -Ui.SafeBottom - 40);
        under.rectTransform.offsetMax = new Vector2(0, -40);
        under.raycastTarget = true;
        var plank = Ui.Skin(bar, "h_tabbar", new Vector2(1100, TabH + 20), new Vector2(0, 4),...
        plank.raycastTarget = true;
        string[] names = { "商城", "好汉", "出征", "寨主", "山寨" };
        string[] icons = { "tab_shop", "tab_hero", "tab_battle", "tab_forge", "tab_fort" };
        bool[] dots = { ShopBadge(), HeroBadge(), BattleBadge(), ForgeBadge(), FortBadge() };
        for (int i = 0; i < 5; i++)
        {
            float x = (i - 2) * 212;
            bool on = i == _tab;
            var tab = Ui.Rect(bar, "tab" + i);
            tab.anchoredPosition = new Vector2(x, 0);
            tab.sizeDelta = new Vector2(212, TabH);
            var face = Ui.Box(tab, new Vector2(204, TabH), Vector2.zero, new Color(0, 0, 0, 0...
            face.raycastTarget = true;
            if (on)
            {
                Ui.Box(tab, new Vector2(260, 200), new Vector2(0, 36), Pal.Gold.A(0.45f), Art...
                Ui.Icon(tab, "ui/h_tab_on", 196, new Vector2(0, 30), "badge");
            }
```

```
            var ic = Ui.Icon(tab, icons[i], on ? 132 : 98, new Vector2(0, on ? 46 : 16));
            if (on) Tween.Run(ic, 999f, k => ic.rectTransform.anchoredPosition = new Vector2(...
            Ui.Outlined(tab, names[i], on ? 30 : 26, on ? Pal.Hex("FFE08A") : Pal.Hex("FBE6C4...
            if (dots[i]) MetaUi.Dot(tab, new Vector2(56, on ? 92 : 50));
            int ti = i;
            Ui.Tap(face, () => SetTab(ti), tab, false);
        }
    }
    static bool ShopBadge() => S.AdIngots < Meta.AdIngotsMax || S.AdRecruit < Meta.AdRecruitM...
    static bool HeroBadge()
    {
        for (int h = 0; h < Defs.HeroCount; h++) if (Meta.CanLevel(S, (HeroId)h)) return true;
        return false;
    }
    static bool ForgeBadge()
    {
        for (int i = 0; i < Meta.Equips.Length; i++) if (Meta.CanForge(S, i)) return true;
        return Meta.CanTalent(S) || Meta.CanPage(S) || Meta.CanTome(S) || Meta.CanFlagUp(S) |...
    }
    static bool IdleBadge() => Meta.IdleHours(S, Profile.Now) >= 1;
    static bool MealBadge() => Meta.MealsReady(S, DateTime.Now.Hour) > 0;
    static bool ChestBadge(int ch)
    {
        for (int k = 0; k < 3; k++) if (Meta.CanOpenChest(S, ch, k)) return true;
        return false;
    }
    static bool BattleBadge()
    {
        if (Meta.CanSign(S, Profile.Today) || Meta.TaskBadge(S) > 0 || Meta.CodexBadge(S) > 0...
        for (int c = 1; c <= Defs.Chapters.Length; c++) if (ChestBadge(c)) return true;
        return false;
    }
    static bool FortBadge() => IdleBadge() || Meta.TaskBadge(S) > 0;
    // ---------- 出征 ----------
    /// <summary>
    /// 参考视频的主页就是战场：城门口站着本关首领，中间是关卡水印，寨墙上站着出战的四位好汉，
    /// 下面一排三个关卡宝箱，再下面是开始挑战。功能入口收进两侧半透明竖条和顶部小图标。
    /// </summary>
    void BuildBattle(bool anim)
    {
        var st = Defs.Level(S.Selected);
        Cover("home_road", new Vector2(1080, 1920), Color.white, 0.5f);
        Shade(0f, 0.22f);
        float sp = Ui.Spare;
        int day = Profile.Today;
        // 顶部小图标（资源条下面）
        float iy = Ui.SafeTop + 150;
        MiniBtn(10, iy, "ic_feast", "聚义宴", MealBadge(), () => MetaUi.Meals(Layer));
        MiniBtn(130, iy, "tab_gift", "签到", Meta.CanSign(S, day), () => MetaUi.SignIn(Layer));
        MiniBtn(250, iy, "ic_map", "关卡", false, () => MetaUi.LevelList(Layer));
```

```
        MiniBtn(370, iy, "ic_gear", "设置", false, () => MetaUi.Settings(Layer));
        // 两侧竖排入口
        bool dOpen = S.Cleared >= MetaUi.DailyUnlock, eOpen = S.Cleared >= MetaUi.EscortUnlock;
        float cy = 290 + sp * 0.05f;
        SideBtn(-470, cy, 0, "ic_medal", "挑战", dOpen && S.DailyTries < Meta.DailyTriesMax, () =>
        {
            if (!dOpen) Ui.Toast($"通关第 {MetaUi.DailyUnlock} 关后开放", 200);
            else MetaUi.Daily(Layer);
        }, !dOpen);
        SideBtn(-470, cy, 1, "ic_cart", "生辰纲", eOpen && S.EscortTries < Meta.EscortTriesMax, ...
        {
            if (!eOpen) Ui.Toast($"通关第 {MetaUi.EscortUnlock} 关后开放", 200);
            else MetaUi.Escort(Layer);
        }, !eOpen);
        SideBtn(-470, cy, 2, "tab_idle", "巡山", IdleBadge(), () => MetaUi.Idle(Layer));
        SideBtn(470, cy, 0, "tab_task", "任务", Meta.TaskBadge(S) > 0, () => MetaUi.Tasks(Layer));
        SideBtn(470, cy, 1, "tab_codex", "图鉴", Meta.CodexBadge(S) > 0, () => MetaUi.Codex(Lay...
        // 首领 + 关卡名
        float by = 330 + sp * 0.1f;
        var bd = Defs.Enemy(st.BossKind);
        var boss = Top(0, by, new Vector2(360, 360), "boss");
        Ui.Box(boss, new Vector2(300, 80), new Vector2(0, -130), new Color(0, 0, 0, 0.3f), Ar...
        Ui.Box(boss, new Vector2(340, 110), new Vector2(0, -128), (st.IsChapterBoss ? Pal.Red...
        float bsz = st.IsChapterBoss ? 330 : 290;
        var icon = Ui.Icon(boss, bd.Art, bsz, new Vector2(0, 0));
        Tween.Run(icon, 999f, k => icon.rectTransform.anchoredPosition = new Vector2(0, 5 * M...
        if (anim) Tween.Pop(boss, 0.35f);
        var tag = Top(0, by + 170, new Vector2(620, 70), "levelTag");
        var nameT = Ui.Outlined(tag, $"{st.No}.{st.Name}", 38, Color.white, new Vector2(24, 4...
        float tw = Mathf.Min(460, nameT.preferredWidth);
        var ribbon = Ui.Skin(tag, "h_ribbon", new Vector2(tw + 190, 72), new Vector2(24, 0), ...
        ribbon.transform.SetAsFirstSibling();
        var list = Ui.Icon(tag, "ui/h_round", 70, new Vector2(24 - tw * 0.5f - 112, 2), "list");
        for (int i = 0; i < 3; i++) Ui.Box(list.transform, new Vector2(28, 5), new Vector2(0,...
        Ui.Tap(list, () => MetaUi.LevelList(Layer), tag);
        // 关卡水印
        var wmEdge = Pal.Hex("6E3A12").A(0.55f);
        float wy = by + 330 + sp * 0.12f;
        var wm = Top(0, wy, new Vector2(1000, 220), "watermark");
        Ui.Outlined(wm, $"第 {st.No} 关", 64, Color.white.A(0.85f), new Vector2(0, 70), new Vec...
        Ui.Outlined(wm, st.Name, st.Name.Length > 4 ? 100 : 116, Color.white.A(0.8f), new Vec...
        int left = Meta.LevelChestLeft(S, st.No);
        var sub = Top(0, wy + 128, new Vector2(560, 50), "chestLeft");
        Ui.Skin(sub, "h_plaque", new Vector2(540, 50), Vector2.zero, -1f);
        Ui.Outlined(sub, left > 0 ? $"本关仍有 <color=#FFD966>{left}</color> 个宝箱未获得" : "本关宝箱已全部获得...
        string elite = st.EliteWaves.Length > 0 ? " · 精英 " + Defs.Enemy(st.EliteKinds[0]).Nam...
        string buff = st.BuffRole.HasValue ? $"  <color=#B6FF8A>增益：{st.BuffText}</color>" : "";
        Ui.Outlined(Top(0, wy + 178, new Vector2(900, 36), "info"), $"{st.Waves} 波 · 首领 {bd.N...
        // 寨墙 + 出战好汉
        float baseY = TabH;
```

```
        float wallY = baseY + 452 + sp * 0.3f;
        var podium = Bottom(0, wallY, new Vector2(1200, 277), "podium");
        var wall = Ui.Icon(podium, "wall", 1200, Vector2.zero, "wall");
        wall.rectTransform.sizeDelta = new Vector2(1200, 277);
        var lu = Meta.Lineup(S);
        for (int i = 0; i < lu.Length; i++)
        {
            var d = Defs.Hero(lu[i]);
            var g = Ui.Rect(podium, "hero" + i);
            g.anchoredPosition = new Vector2((i - 1.5f) * 196, 96);
            g.sizeDelta = new Vector2(190, 200);
            Ui.Box(g, new Vector2(190, 70), new Vector2(0, -62), Pal.Teal.A(0.55f), Art.SoftD...
            Ui.Icon(g, "ui/h_pedestal", 150, new Vector2(0, -76), "pedestal");
            var art = Ui.Icon(g, d.Art, 186, new Vector2(0, 18), "art");
            float ph = i * 0.8f;
            Tween.Run(art, 999f, k => art.rectTransform.anchoredPosition = new Vector2(0, 18 ...
            Ui.Tap(art, () => SetTab(TabHero), g);
            if (Meta.CanLevel(S, lu[i])) MetaUi.Dot(g, new Vector2(66, 92));
            if (anim) Tween.Pop(g, 0.3f, 0.1f + i * 0.05f);
        }
        var pw = Ui.Rect(podium, "power");
        pw.anchoredPosition = new Vector2(0, -60);
        pw.sizeDelta = new Vector2(250, 56);
        Ui.Skin(pw, "h_plaque", new Vector2(250, 56), Vector2.zero, -1f);
        Ui.Icon(pw, "ic_sword", 48, new Vector2(-82, 2));
        Ui.Outlined(pw, Ui.Num(Meta.TeamPower(S)), 32, Pal.Hex("FFE08A"), new Vector2(16, 1),...
        // 关卡三宝箱
        float chestY = baseY + 262 + sp * 0.2f;
        var row = Bottom(0, chestY, new Vector2(480, 130), "chests");
        Ui.Skin(row, "h_capsule", new Vector2(460, 108), new Vector2(0, -6), -1f);
        for (int k = 0; k < 3; k++)
        {
            bool got = Meta.LevelChestGot(S, st.No, k);
            float x = (k - 1) * 140;
            var c = Ui.Icon(row, got ? "ic_chest_open" : MetaUi.ChestArt(k), 116, new Vector2...
            if (got) MetaUi.Check(row, new Vector2(x + 38, -30), 34);
            int ck = k, lvl = st.No;
            Ui.Tap(c, () => MetaUi.LevelChestCard(Layer, lvl, ck, c.transform));
            if (anim) Tween.Pop(c.transform, 0.28f, 0.15f + k * 0.05f);
        }
        // 左：章节星级宝箱；右：扫荡
        int stars = Meta.ChapterStars(S, st.Chapter);
        var cs = Bottom(-410, chestY, new Vector2(170, 120), "chapterStars");
        var csBg = Ui.Skin(cs, "h_tile", new Vector2(160, 116), Vector2.zero, 0.5f);
        Ui.Icon(cs, "ic_chest_gold", 74, new Vector2(-34, 10));
        Ui.Icon(cs, "ic_star", 36, new Vector2(38, 22));
        Ui.Outlined(cs, $"{stars}/30", 24, Color.white, new Vector2(30, -28), new Vector2(110...
        if (ChestBadge(st.Chapter)) MetaUi.Dot(cs, new Vector2(70, 46));
        int chap = st.Chapter;
        Ui.Tap(csBg, () => MetaUi.ChapterChests(Layer, chap), cs);
```

```
        if (Meta.CanSweep(S, st.No))
        {
            var sw = Bottom(410, chestY, new Vector2(150, 150), "sweep");
            var swBg = Ui.Icon(sw, "ui/h_round", 116, new Vector2(0, 14), "bg");
            Ui.Icon(sw, "ic_sweep", 82, new Vector2(0, 16));
            Ui.Outlined(sw, "扫荡", 28, Color.white, new Vector2(0, -44), new Vector2(150, 36),...
            Ui.Outlined(sw, $"体力 -{Meta.BattleStamina}", 20, Pal.Cream, new Vector2(0, -72), ...
            Ui.Tap(swBg, () => Sweep(st), sw);
        }
        // 开始挑战：朱金大匾，字画在面上，按下整块下沉。
        float startY = baseY + 112 + sp * 0.12f;
        const float bw = 560f, bh = 150f;
        var start = Ui.Rect(_page, "start");
        start.anchorMin = start.anchorMax = new Vector2(0.5f, 0);
        start.sizeDelta = new Vector2(bw, bh);
        start.anchoredPosition = new Vector2(0, startY);
        var body = Ui.Rect(start, "body");
        body.sizeDelta = new Vector2(bw, bh);
        var face = Ui.Skin(body, "h_start", new Vector2(bw, bh), Vector2.zero, 0.55f, null, "...
        face.raycastTarget = false;
        Ui.Outlined(body, "开始挑战", 50, Color.white, new Vector2(0, 20), new Vector2(bw - 90, 6...
        Ui.Icon(body, "ic_wine", 38, new Vector2(-18, -32));
        Ui.Outlined(body, Meta.BattleStamina.ToString(), 28, Color.white, new Vector2(16, -31...
        var hit = Ui.Box(start, new Vector2(bw, bh), Vector2.zero, new Color(0, 0, 0, 0), Art...
        Ui.Tap(hit, () => Game.I.StartLevel(S.Selected), body, true, 8f);
        var glow = Ui.Box(start, new Vector2(bw + 80, bh + 50), Vector2.zero, Pal.Gold.A(0.22...
        glow.transform.SetAsFirstSibling();
        Tween.Run(start, 9999f, k =>
        {
            float t = k * 9999f;
            start.localScale = Vector3.one * (1f + 0.025f * Mathf.Sin(t * 4f));
            glow.color = Pal.Gold.A(0.14f + 0.1f * Mathf.Sin(t * 4f));
        });
        float ax = bw * 0.5f + 78f;
        if (S.Selected > 1) ArrowBtn(-ax, startY, -1);
        if (S.Selected <= S.Cleared && S.Selected < Defs.LevelCount) ArrowBtn(ax, startY, 1);
    }
    void Sweep(LevelDef st)
    {
        if (S.Stamina < Meta.BattleStamina)
        {
            Sfx.Play(Snd.Deny);
            MetaUi.StaminaShop(Layer);
            return;
        }
        var r = Meta.Sweep(S, st, new Rng(Environment.TickCount));
        if (r == null) return;
        Profile.Save();
        Sfx.Play(Snd.Coin);
        MetaUi.ShowRewards(Layer, $"扫荡 第{st.No}关", r);
```

```
    }
    void ArrowBtn(float x, float y, int dir)
    {
        var b = Ui.Icon(_page, "ui/h_arrow", 100, Vector2.zero, "arrow");
        b.rectTransform.anchorMin = b.rectTransform.anchorMax = new Vector2(0.5f, 0);
        b.rectTransform.anchoredPosition = new Vector2(x, y);
        if (dir > 0) b.rectTransform.localScale = new Vector3(-1, 1, 1);
        Ui.Tap(b, () =>
        {
            S.Selected = Mathf.Clamp(S.Selected + dir, 1, Mathf.Min(Defs.LevelCount, S.Cleare...
            Profile.Save();
        });
    }
    const float SideStep = 146;
    /// <summary>两侧竖排入口：金边朱漆方块里放图标，字压在方块下沿。</summary>
    void SideBtn(float x, float y, int idx, string icon, string name, bool dot, Action act, b...
    {
        var b = Top(x, y + idx * SideStep, new Vector2(120, 136), "side_" + name);
        var tile = Ui.Skin(b, "h_tile", new Vector2(112, 112), new Vector2(0, 8), 0.5f, locke...
        var ic = Ui.Icon(b, icon, 92, new Vector2(0, 12));
        if (locked)
        {
            ic.color = new Color(0.55f, 0.55f, 0.6f);
            Ui.Icon(b, "ic_lock", 40, new Vector2(32, -16));
        }
        Ui.Outlined(b, name, 26, Color.white, new Vector2(0, -50), new Vector2(140, 34), Pal....
        if (dot) MetaUi.Dot(b, new Vector2(46, 56)).rectTransform.localScale = Vector3.one * ...
        Ui.Tap(tile, act, b);
    }
    /// <summary>资源条下面一排小入口：金圈红底圆钮里放图标，下面小字。</summary>
    void MiniBtn(float x, float y, string icon, string name, bool dot, Action act)
    {
        var b = Ui.Top(_page, x, y, new Vector2(112, 112), "mini_" + name);
        var ring = Ui.Icon(b, "ui/h_round", 88, new Vector2(0, 14), "ring");
        Ui.Icon(b, icon, 70, new Vector2(0, 16));
        Ui.Outlined(b, name, 23, Color.white, new Vector2(0, -38), new Vector2(124, 30), Pal....
        if (dot) MetaUi.Dot(b, new Vector2(34, 48)).rectTransform.localScale = Vector3.one * ...
        Ui.Tap(ring, act, b);
    }
    // ---------- 好汉 ----------
    /// <summary>换人模式：详情里点了「上阵」的好汉，等玩家点顶上要替换的位置。</summary>
    HeroId? _swap;
    int _popSlot = -1;
    public void BeginSwap(HeroId h)
    {
        _swap = h;
        Sfx.Play(Snd.Swap, 0.6f);
        Rebuild(false);
    }
    void EndSwap(int slot)
```

```
    {
        if (_swap.HasValue && slot >= 0)
        {
            var h = _swap.Value;
            Meta.SetLineup(S, slot, h);
            Profile.Save();
            Sfx.Play(Snd.Swap);
            Ui.Toast($"{Defs.Hero(h).Name} 上阵", 200);
            _popSlot = slot;
        }
        _swap = null;
        Rebuild(false);
    }
    void BuildHeroes(bool anim)
    {
        Cover("hall_bg", new Vector2(1080, 1920), Color.white, 0.3f);
        Shade(0.3f, 0f);
        float sp = Ui.Spare;
        // 上半：出战的四位站在场景里，下方三个阵容方案
        var line = Meta.Lineup(S);
        float luY = 420 + sp * 0.16f;
        var lu = Top(0, luY, new Vector2(1000, 300), "lineup");
        lu.localScale = Vector3.one * (1f + Mathf.Min(0.1f, sp / 2500f));
        for (int i = 0; i < 4; i++)
        {
            var h = line[i];
            var d = Defs.Hero(h);
            var g = Ui.Rect(lu, "lu" + i);
            g.anchoredPosition = new Vector2((i - 1.5f) * 250, 0);
            g.sizeDelta = new Vector2(240, 300);
            Ui.Box(g, new Vector2(230, 80), new Vector2(0, -104), MetaUi.RarityColor(d.Rarity...
            Ui.Box(g, new Vector2(190, 52), new Vector2(0, -108), new Color(0, 0, 0, 0.35f), ...
            var art = Ui.Icon(g, d.Art, 250, new Vector2(0, 14), "art");
            float ph = i * 0.7f;
            Tween.Run(art, 999f, k => art.rectTransform.anchoredPosition = new Vector2(0, 14 ...
            var plate = Ui.Skin(g, "plate", new Vector2(206, 50), new Vector2(0, -128), -1f, ...
            Ui.Outlined(plate.transform, $"Lv.{S.HeroLv[(int)h]} {d.Name}", 25, Color.white, ...
            int slot = i;
            if (_swap.HasValue)
            {
                var arrow = Ui.Icon(g, "ui/arrow", 86, new Vector2(0, 160), "swapArrow");
                arrow.transform.localRotation = Quaternion.Euler(0, 0, 90);
                arrow.color = Pal.Hex("8CFF6A");
                float off = i * 0.4f;
                Tween.Run(arrow, 999f, k => arrow.rectTransform.anchoredPosition = new Vector...
                Ui.Tap(art, () => EndSwap(slot), g);
                Ui.Tap(arrow, () => EndSwap(slot), g);
            }
            else
            {
```

```
            Ui.Tap(art, () => MetaUi.HeroDetail(Layer, h), g);
            if (Meta.CanLevel(S, h)) MetaUi.Dot(g, new Vector2(98, -104));
        }
        if (anim) Tween.Pop(g, 0.3f, 0.05f + i * 0.05f);
        if (_popSlot == i)
        {
            Tween.Pop(g, 0.35f);
            MetaFx.Burst(lu, g.anchoredPosition, 10, 180);
        }
    }
    _popSlot = -1;
    float pgY = luY + 200;
    for (int i = 0; i < Meta.PresetCount; i++)
    {
        bool on = i == S.Preset;
        var dot = Top((i - 1) * 96, pgY, new Vector2(80, 80), "preset" + i);
        var face = Ui.Box(dot, Vector2.one * (on ? 72 : 60), Vector2.zero, on ? Pal.Gold ...
        Ui.Box(dot, Vector2.one * (on ? 60 : 50), Vector2.zero, on ? Pal.Hex("FFE48A") : ...
        Ui.Outlined(dot, (i + 1).ToString(), on ? 36 : 30, on ? Pal.Hex("7A4A08") : Pal.H...
        int pi = i;
        Ui.Tap(face, () =>
        {
            if (pi == S.Preset) return;
            Meta.UsePreset(S, pi);
            Profile.Save();
            Sfx.Play(Snd.Swap);
            Ui.Toast($"阵容方案 {pi + 1}", 200);
            _swap = null;
            Rebuild(false);
            var l = _page.Find("body/lineup");
            if (l) for (int k = 0; k < l.childCount; k++) Tween.Pop(l.GetChild(k), 0.28f,...
        }, dot);
    }
    var pw = Top(0, pgY + 86, new Vector2(420, 76), "power");
    Ui.Skin(pw, "plate", new Vector2(380, 70), Vector2.zero, -1f);
    Ui.Icon(pw, "ic_sword", 70, new Vector2(-120, 2));
    Ui.Outlined(pw, Ui.Num(Meta.TeamPower(S)), 40, Color.white, new Vector2(24, 2), new V...
    var cd = Top(0, pgY + 160, new Vector2(820, 64), "critBand");
    Ui.Skin(cd, "pill", new Vector2(760, 62), Vector2.zero, -1f);
    Ui.Outlined(cd, $"暴击伤害增幅：<color=#FFC93C>{Meta.BuildLoadout(S).CritDmg * 100:0.#}%</co...
    var tip = Ui.Box(cd, Vector2.one * 46, new Vector2(330, 0), Pal.Hex("E6EEF6"), Art.Ci...
    Ui.Outlined(tip.transform, "!", 32, Pal.Navy, new Vector2(0, 2), new Vector2(46, 46),...
    Ui.Tap(tip, () => Ui.Toast("任意好汉每升 1 级，全军暴击伤害 +1%", 200));
    // 下半：深色面板里的好汉卡
    float py = pgY + 210;
    var panel = Top(0, py, new Vector2(1060, 10), "roster");
    panel.pivot = new Vector2(0.5f, 1f);
    float ph2 = Mathf.Max(820f, Ui.Height - Ui.SafeTop - py - 176f);
    Ui.Skin(panel, "dark", new Vector2(1060, ph2), new Vector2(0, -ph2 * 0.5f), 1f, Pal.H...
    var roles = new List<int> { -1 };
```

```
        foreach (var r in Defs.RoleOrder) roles.Add((int)r);
        int owned = 0;
        for (int h = 0; h < Defs.HeroCount; h++) if (Meta.Owned(S, (HeroId)h)) owned++;
        for (int i = 0; i < roles.Count; i++)
        {
            int rf = roles[i];
            bool sel = rf == _roleFilter;
            var col = rf < 0 ? Pal.Hex("C89A4A") : MetaUi.RoleColor((Role)rf);
            var pos = new Vector2((i - (roles.Count - 1) * 0.5f) * 144, -54);
            var tb = Ui.Skin(panel, sel ? Ui.BtnSkin(col) : "row", new Vector2(136, 64), pos,...
            if (rf >= 0) Ui.Icon(tb.transform, MetaUi.RoleArt((Role)rf), 38, new Vector2(-40,...
            Ui.Outlined(tb.transform, rf < 0 ? $"全部{owned}" : Defs.RoleName((Role)rf), rf < 0...
            Ui.Tap(tb, () =>
            {
                if (_roleFilter == rf) return;
                _roleFilter = rf;
                _rosterScroll = 0;
                Sfx.Play(Snd.Tab, 0.6f);
                Rebuild(false);
            }, tb.transform, false);
        }
        var order = new List<HeroId>();
        for (int h = 0; h < Defs.HeroCount; h++)
            if (_roleFilter < 0 || (int)Defs.Hero((HeroId)h).Kind == _roleFilter) order.Add((...
        order.Sort((a, b) =>
        {
            bool oa = Meta.Owned(S, a), ob = Meta.Owned(S, b);
            if (oa != ob) return oa ? -1 : 1;
            bool la = Array.IndexOf(S.Lineup, (int)a) >= 0, lb = Array.IndexOf(S.Lineup, (int...
            if (la != lb) return la ? -1 : 1;
            int r = Defs.Hero(b).Rarity.CompareTo(Defs.Hero(a).Rarity);
            if (r != 0) return r;
            if (!oa)
            {
                int ua = Meta.UnlockLevelOf(a), ub = Meta.UnlockLevelOf(b);
                if (ua != ub) return (ua == 0 ? 999 : ua).CompareTo(ub == 0 ? 999 : ub);
            }
            return S.HeroLv[(int)b] != S.HeroLv[(int)a] ? S.HeroLv[(int)b].CompareTo(S.HeroLv...
        });
        float viewH = ph2 - 106;
        var view = Ui.Rect(panel, "view");
        view.anchorMin = view.anchorMax = new Vector2(0.5f, 1f);
        view.pivot = new Vector2(0.5f, 1f);
        view.sizeDelta = new Vector2(1030, viewH);
        view.anchoredPosition = new Vector2(0, -96);
        view.gameObject.AddComponent<RectMask2D>();
        Ui.Fill(view, new Color(0, 0, 0, 0.001f), true, "hit");
        var content = Ui.Rect(view, "content");
        content.anchorMin = content.anchorMax = new Vector2(0.5f, 1f);
        content.pivot = new Vector2(0.5f, 1f);
```

```
        const float rowH = 318;
        int rows = (order.Count + 3) / 4;
        content.sizeDelta = new Vector2(1030, rows * rowH + 80);
        var top = Ui.Rect(content, "top");
        top.anchorMin = top.anchorMax = new Vector2(0.5f, 1f);
        top.sizeDelta = Vector2.zero;
        for (int i = 0; i < order.Count; i++)
        {
            var c = HeroCard(top, order[i], new Vector2(((i % 4) - 1.5f) * 250, -(162 + (i / ...
            if (anim && i < 12) Tween.Pop(c, 0.28f, 0.04f + i * 0.025f);
        }
        Ui.Outlined(top, "碎片来源：通关 · 擂鼓招募 · 签到 · 章节宝箱", 26, Pal.Cream, new Vector2(0, -(rows *...
        var sr = view.gameObject.AddComponent<ScrollRect>();
        sr.content = content;
        sr.horizontal = false;
        sr.movementType = ScrollRect.MovementType.Elastic;
        sr.scrollSensitivity = 30;
        content.anchoredPosition = new Vector2(0, Mathf.Clamp(_rosterScroll, 0, Mathf.Max(0, ...
        sr.onValueChanged.AddListener(_ => _rosterScroll = content.anchoredPosition.y);
        if (_swap.HasValue)
        {
            var sh = _swap.Value;
            var dim = Ui.Box(panel, new Vector2(1060, ph2), new Vector2(0, -ph2 * 0.5f), new ...
            dim.raycastTarget = true;
            Ui.Tap(dim, () => EndSwap(-1), null, false);
            var card = HeroCard(panel, sh, new Vector2(0, -ph2 * 0.36f));
            card.localScale = Vector3.one * 1.3f;
            foreach (var g in card.GetComponentsInChildren<Graphic>()) g.raycastTarget = false;
            var glow = Ui.Box(panel, new Vector2(520, 560), new Vector2(0, -ph2 * 0.36f), Pal...
            glow.transform.SetSiblingIndex(card.GetSiblingIndex());
            Ui.Outlined(panel, "点击上方要替换的好汉", 40, Pal.Hex("FFE08A"), new Vector2(0, -ph2 * 0.3...
            Ui.Btn(panel, "取消", new Vector2(240, 96), new Vector2(0, -ph2 * 0.36f - 360), Pal...
        }
    }
    /// <summary>好汉卡：职业圆标、出战中、等级圈、碎片条、可升级绿箭头；未获得的压暗加锁并写来源。</summary
    RectTransform HeroCard(Transform parent, HeroId h, Vector2 pos)
    {
        var d = Defs.Hero(h);
        int lv = S.HeroLv[(int)h];
        bool owned = lv > 0;
        var c = Ui.Rect(parent, "card_" + h);
        c.anchoredPosition = pos;
        c.sizeDelta = new Vector2(236, 300);
        var face = MetaUi.Card(c, owned ? MetaUi.RarityCard(d.Rarity) : "card_lock", new Vect...
        var art = Ui.Icon(c, d.Art, 216, new Vector2(0, 44));
        if (!owned) art.color = new Color(0.1f, 0.1f, 0.13f, 0.85f);
        Ui.Box(c, Vector2.one * 58, new Vector2(-84, 120), Color.white, Art.Circle, "roleRing");
        Ui.Box(c, Vector2.one * 50, new Vector2(-84, 120), owned ? MetaUi.RoleColor(d.Kind) :...
        Ui.Icon(c, MetaUi.RoleArt(d.Kind), 38, new Vector2(-84, 121));
        Ui.Outlined(c, d.Name, 30, owned ? Color.white : Pal.Hex("B8C0CC"), new Vector2(0, -1...
```

```
        if (owned)
        {
            if (Array.IndexOf(S.Lineup, (int)h) >= 0)
            {
                var on = Ui.Icon(c, "ui/tag", 118, new Vector2(-54, -30), "onTag");
                Ui.Outlined(on.transform, "出战中", 24, Color.white, new Vector2(-6, 2), new Vec...
            }
            var lvc = Ui.Icon(c, "ui/lv", 60, new Vector2(-86, -76), "lv");
            Ui.Outlined(lvc.transform, lv.ToString(), 28, Pal.Hex("FFE7B0"), new Vector2(0, 1...
            int need = Meta.ShardCost(lv);
            bool max = lv >= Meta.MaxHeroLv;
            bool can = Meta.CanLevel(S, h);
            MetaUi.Bar(c, new Vector2(16, -76), new Vector2(128, 26), max ? 1 : S.Shards[(int...
                max ? "满级" : $"{S.Shards[(int)h]}/{need}");
            if (can)
            {
                var up = Ui.Icon(c, "ic_fast", 46, new Vector2(98, -70), "up");
                up.transform.localRotation = Quaternion.Euler(0, 0, 90);
                up.color = Pal.Hex("7CFF5A");
                Tween.Run(up, 999f, k => up.rectTransform.anchoredPosition = new Vector2(98, ...
                MetaUi.Dot(c, new Vector2(100, 136));
            }
        }
        else
        {
            Ui.Icon(c, "ic_lock", 50, new Vector2(84, 120));
            int at = Meta.UnlockLevelOf(h);
            string src = at > 0 ? $"第{at}关入伙" : d.Rarity >= 4 && !Meta.OrangeOpened(S) ? $"第{...
            Ui.Outlined(c, src, 24, Pal.Hex("D8D0C0"), new Vector2(0, -76), new Vector2(220, ...
        }
        Ui.Tap(face, () => MetaUi.HeroDetail(Layer, h), c);
        return c;
    }
    // ---------- 寨主（装备） ----------
    void BuildForge(bool anim)
    {
        Cover("forge_bg", new Vector2(1080, 1920), Color.white, 0.5f);
        Shade(0.3f, 0.25f);
        float sp = Ui.Spare;
        var hdr = Top(0, 228 + sp * 0.05f, new Vector2(700, 110), "hdr");
        Ui.Ribbon(hdr, "寨主 · 宋江", Vector2.zero, 520);
        var lo = Meta.BuildLoadout(S);
        var st = Top(0, 340 + sp * 0.08f, new Vector2(980, 80), "stats");
        Ui.Skin(st, "pill", new Vector2(960, 70), Vector2.zero, -1f);
        Ui.Icon(st, "ic_sword", 54, new Vector2(-420, 0));
        Ui.Outlined(st, $"攻击力 {lo.Atk:0}　暴击 {lo.Crit * 100:0.#}%　暴伤 {lo.CritDmg * 100:0...
        var sk = Top(0, 418 + sp * 0.1f, new Vector2(580, 60), "lordSkill");
        var skBg = Ui.Box(sk, new Vector2(580, 58), Vector2.zero, Pal.Hex("3A2410").A(0.85f),...
        Ui.Icon(sk, "ic_sword", 46, new Vector2(-252, 0)).color = Pal.Gold;
        Ui.Outlined(sk, $"{Defs.LordSkillName}：每{lo.LordCd:0.#}秒 · {lo.LordPct * 100:0}%攻击力",...
```

```
        Ui.Tap(skBg, () => Ui.Toast($"天赋每10层 +{Meta.LordPctStep * 100:0}% 伤害，号角减冷却", 360), sk);
        var sj = Top(0, 1000 + sp * 0.4f, new Vector2(640, 640), "songjiang");
        sj.localScale = Vector3.one * (1f + Mathf.Min(0.2f, sp / 2000f));
        Ui.Box(sj, new Vector2(700, 380), new Vector2(0, -200), Pal.Gold.A(0.35f), Art.SoftDot);
        Ui.Box(sj, new Vector2(400, 90), new Vector2(0, -290), new Color(0, 0, 0, 0.3f), Art....
        var art = Ui.Icon(sj, "prop_songjiang", 640, Vector2.zero);
        Tween.Run(art, 999f, k => art.rectTransform.anchoredPosition = new Vector2(0, 5 * Mat...
        if (anim) Tween.Pop(sj, 0.35f);
        for (int i = 0; i < Meta.Equips.Length; i++)
        {
            float x = i < 3 ? -390 : 390;
            float y = 500 + sp * 0.22f + (i % 3) * (270 + sp * 0.17f);
            var e = Meta.Equips[i];
            int lv = S.EquipLv[i];
            int q = S.EquipQ[i];
            var c = Top(x, y, new Vector2(220, 230), "eq" + i);
            var face = MetaUi.Slot(c, MetaUi.QualitySlot(q), 196, new Vector2(0, 26));
            Ui.Icon(c, e.Icon, 150, new Vector2(0, 30));
            var lvb = Ui.Skin(c, "pill", new Vector2(116, 44), new Vector2(46, -50), -1f);
            Ui.Outlined(lvb.transform, "Lv." + lv, 26, Color.white, new Vector2(0, 1), new Ve...
            if (q > 0) Ui.Outlined(c, Meta.QualityName[q], 22, MetaUi.QualityColor(q), new Ve...
            Ui.Outlined(c, e.Name, 30, Color.white, new Vector2(0, -96), new Vector2(220, 44)...
            if (Meta.CanForge(S, i)) MetaUi.Dot(c, new Vector2(88, 112));
            int si = i;
            Ui.Tap(face, () => MetaUi.ForgeSlot(Layer, si), c);
            if (anim) Tween.Pop(c, 0.3f, 0.1f + i * 0.05f);
        }
        var sum = Bottom(0, 440 + sp * 0.12f, new Vector2(980, 300), "summary");
        Ui.Panel(sum, new Vector2(940, 280), Vector2.zero);
        for (int i = 0; i < Meta.Equips.Length; i++)
        {
            int lv = S.EquipLv[i];
            string brk = Meta.Breaks(lv) > 0 ? $" <color=#2E9A38>{Defs.RoleName(Meta.Equips[i...
            Ui.Label(sum, Meta.EquipText(i, lv, S.EquipQ[i]) + brk, 26, Pal.Ink, new Vector2(...
        }
        Ui.Label(sum, $"强化上限 Lv.{Meta.EquipMax(S)}（随通关提升）", 26, Pal.Hex("8A4A1A"), new Vector...
        MetaUi.Pill(sum, "ic_iron", Ui.Num(S.Iron), new Vector2(320, -88), 220);
        float ey = 640 + sp * 0.16f;
        Action re = () => Rebuild(false);
        ForgeEntry(-375, ey, "ic_scroll", "天赋", $"{S.Talent}/{Meta.TalentCap(S)}", Meta.CanTa...
        ForgeEntry(-125, ey, "ic_tome", "天书", $"上卷 Lv.{S.PageLv}", Meta.CanPage(S) || Meta.Ca...
        ForgeEntry(125, ey, "ic_flag", "杏黄旗", S.FlagLv > 0 ? $"Lv.{S.FlagLv}" : "未打造", Meta.C...
        ForgeEntry(375, ey, "ic_gearbox", "装备箱", $"×{S.GearBoxes}", S.GearBoxes > 0, () => Me...
    }
    void ForgeEntry(float x, float y, string icon, string name, string sub, bool dot, Action ...
    {
        var b = Bottom(x, y, new Vector2(236, 120), "entry_" + name);
        var bg = Ui.Box(b, new Vector2(236, 116), Vector2.zero, Glass.A(0.82f), Art.Rounded, ...
        Ui.Box(b, new Vector2(230, 110), Vector2.zero, Pal.Hex("2A4466").A(0.6f), Art.Rounded...
        Ui.Icon(b, icon, 84, new Vector2(-66, 2));
```

```
12458            if (_rt != null) _pos = _rt.anchoredPosition;
12459            _tints.Clear();
12460            foreach (var g in _t.GetComponentsInChildren<Graphic>(false))
12461            {
12462               if (g.color.a < 0.2f) continue;
12463               var n = g.gameObject.name;
12464               if (n == "glow" || n == "hit" || n == "lip") continue;
12465               _tints.Add(new Tint { G = g, C = g.color });
12466            }
12467         }
12468         Apply(true);
12469      }
12470      public void OnPointerEnter(PointerEventData e)
12471      {
12472         _over = true;
12473         if (_down) Apply(true);
12474      }
12475      public void OnPointerExit(PointerEventData e)
12476      {
12477         _over = false;
12478         if (_sunk) Release(false);
12479      }
12480      public void OnPointerUp(PointerEventData e)
12481      {
12482         bool back = _down && _over;
12483         _down = false;
12484         if (_sunk) Release(back);
12485      }
12486      void LateUpdate()
12487      {
12488         if (!_down) return;
12489         if (Input.GetMouseButton(0) || Input.touchCount > 0) return;
12490         _down = false;
12491         if (_sunk) Release(false);
12492      }
12493      void OnDisable()
12494      {
12495         _down = false;
12496         if (_sunk && _t != null) Restore();
12497         _sunk = false;
12498      }
12499      void Apply(bool sunk)
12500      {
12501         if (_t == null || TooBig || !_have) return;
12502         _sunk = sunk;
12503         float k = sunk ? (_sink > 0f ? 0.985f : 0.93f) : 1f;
12504         _t.localScale = new Vector3(_scale.x * k, _scale.y * k, _scale.z);
12505         if (_rt != null && _sink > 0f)
12506            _rt.anchoredPosition = _pos + (sunk ? new Vector2(0f, -_sink) : Vector2.zero);
12507         foreach (var tn in _tints)
```

```
            if (tn.G) tn.G.color = sunk ? tn.C * 0.82f : tn.C;
        }
        void Release(bool spring)
        {
            _sunk = false;
            if (_t == null || !spring)
            {
                Restore();
                return;
            }
            int gen = _gen;
            var fromS = _t.localScale;
            var fromP = _rt != null ? _rt.anchoredPosition : Vector2.zero;
            foreach (var tn in _tints)
                if (tn.G) tn.G.color = tn.C;
            Tween.Run(_t, 0.14f, k =>
            {
                if (gen != _gen || _t == null) return;
                float e = Tween.OutBack(Mathf.Clamp01(k));
                _t.localScale = Vector3.LerpUnclamped(fromS, _scale, e);
                if (_rt != null) _rt.anchoredPosition = Vector2.LerpUnclamped(fromP, _pos, e);
            }, () =>
            {
                if (gen != _gen || _t == null) return;
                Restore();
            });
        }
        void Restore()
        {
            if (_t == null || !_have) return;
            _t.localScale = _scale;
            if (_rt != null) _rt.anchoredPosition = _pos;
            foreach (var tn in _tints)
                if (tn.G) tn.G.color = tn.C;
            _sunk = false;
        }
    }
}
// ===== FILE: Assets/Scripts/Game/WorldFx.cs =====
using System;
using System.Collections.Generic;
using Liangshan.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Liangshan
{
    /// <summary>战场上的一次性特效：爆炸、光圈、落雷、飞金币、漂字、震屏。全部走对象池，随 timeScale 暂停。</sum
    public sealed class WorldFx : MonoBehaviour
    {
        public static WorldFx I { get; private set; }
```

```
    sealed class Job
    {
        public GameObject Go;
        public float T, Dur, Delay;
        public Action<float> Step;
        public bool Release;
    }
    readonly List<Job> _jobs = new List<Job>();
    readonly List<Job> _add = new List<Job>();
    readonly Stack<SpriteRenderer> _pool = new Stack<SpriteRenderer>();
    readonly Stack<LineRenderer> _lines = new Stack<LineRenderer>();
    readonly Stack<Text> _texts = new Stack<Text>();
    int _liveTexts;
    Material _lineMat;
    RectTransform _textLayer;
    Transform _cam;
    Vector3 _camBase;
    float _shake, _shakeT;
    public int LiveTexts => _liveTexts;
    /// <summary>选卡等弹层打开时藏起飘字，免得数字穿进按钮和说明。</summary>
    public void ShowNumbers(bool on)
    {
        if (_textLayer == null) return;
        var g = _textLayer.GetComponent<CanvasGroup>() ?? _textLayer.gameObject.AddComponent<...
        g.alpha = on ? 1f : 0f;
        g.blocksRaycasts = false;
    }
    public static WorldFx Create(Transform parent, RectTransform textLayer, Transform cam)
    {
        var go = new GameObject("WorldFx");
        go.transform.SetParent(parent, false);
        I = go.AddComponent<WorldFx>();
        I._textLayer = textLayer;
        I._cam = cam;
        I._camBase = cam.localPosition;
        I._lineMat = AddMat;
        return I;
    }
    void OnDestroy()
    {
        if (I == this) I = null;
        if (_cam) _cam.localPosition = _camBase;
    }
    // ---------- 基础件 ----------
    /// <summary>sprite 长边 = 1 单位，所以 size 就是长边的世界长度。</summary>
    public static SpriteRenderer Spr(Transform parent, Sprite s, Vector2 pos, float size, int...
    {
        var go = new GameObject(name);
        go.transform.SetParent(parent, false);
        go.transform.localPosition = pos;
```

```
12608         go.transform.localScale = Vector3.one * size;
12609         var sr = go.AddComponent<SpriteRenderer>();
12610         sr.sprite = s;
12611         sr.sortingOrder = order;
12612         sr.color = c ?? Color.white;
12613         return sr;
12614       }
12615       /// <summary>九宫格块，size 为世界尺寸。</summary>
12616       public static SpriteRenderer Block(Transform parent, Sprite s, Vector2 pos, Vector2 size,...
12617       {
12618         var sr = Spr(parent, s, pos, 1f, order, c, name);
12619         sr.drawMode = SpriteDrawMode.Sliced;
12620         sr.size = size;
12621         return sr;
12622       }
12623       public static int Order(float y) => 5000 - Mathf.RoundToInt(y * 100f);
12624       static Material _addMat, _defMat;
12625       /// <summary>叠加发光材质：亮的地方把底下的地面、人物一起照亮，不再像贴纸浮在上面。</summary>
12626       public static Material AddMat
12627       {
12628         get
12629         {
12630           if (_addMat == null)
12631           {
12632             var sh = Resources.Load<Shader>("Shaders/FxAdd");
12633             _addMat = new Material(sh != null ? sh : Shader.Find("Sprites/Default"));
12634           }
12635           return _addMat;
12636         }
12637       }
12638       public static SpriteRenderer Additive(SpriteRenderer sr)
12639       {
12640         sr.sharedMaterial = AddMat;
12641         return sr;
12642       }
12643       SpriteRenderer Take(Sprite s, Vector2 pos, float size, int order, Color c, bool add = false)
12644       {
12645         SpriteRenderer sr;
12646         if (_pool.Count > 0)
12647         {
12648           sr = _pool.Pop();
12649           sr.gameObject.SetActive(true);
12650         }
12651         else
12652         {
12653           var go = new GameObject("fx");
12654           go.transform.SetParent(transform, false);
12655           sr = go.AddComponent<SpriteRenderer>();
12656           if (_defMat == null) _defMat = sr.sharedMaterial;
12657         }
```

```
12658         sr.sharedMaterial = add ? AddMat : _defMat;
12659         sr.sprite = s;
12660         sr.drawMode = SpriteDrawMode.Simple;
12661         sr.sortingOrder = order;
12662         sr.color = c;
12663         sr.flipX = false;
12664         var t = sr.transform;
12665         t.localPosition = pos;
12666         t.localRotation = Quaternion.identity;
12667         t.localScale = Vector3.one * size;
12668         return sr;
12669     }
12670     public void Anim(GameObject go, float dur, Action<float> step, bool release = true, float...
12671     {
12672         if (delay > 0) go.SetActive(false);
12673         _add.Add(new Job { Go = go, Dur = Mathf.Max(0.0001f, dur), Step = step, Release = rel...
12674     }
12675     void Update()
12676     {
12677         if (_add.Count > 0) { _jobs.AddRange(_add); _add.Clear(); }
12678         float dt = Time.deltaTime;
12679         for (int i = _jobs.Count - 1; i >= 0; i--)
12680         {
12681             var j = _jobs[i];
12682             if (j.Go == null) { _jobs.RemoveAt(i); continue; }
12683             if (j.Delay > 0)
12684             {
12685                 j.Delay -= dt;
12686                 if (j.Delay > 0) continue;
12687                 j.Go.SetActive(true);
12688             }
12689             j.T += dt;
12690             float k = Mathf.Clamp01(j.T / j.Dur);
12691             j.Step(k);
12692             if (k < 1f) continue;
12693             _jobs.RemoveAt(i);
12694             if (j.Release) Release(j.Go);
12695         }
12696         UpdateShake(dt);
12697     }
12698     void Release(GameObject go)
12699     {
12700         if (go.TryGetComponent<SpriteRenderer>(out var sr) && go.transform.parent == transform)
12701         {
12702             go.SetActive(false);
12703             _pool.Push(sr);
12704         }
12705         else if (go.TryGetComponent<LineRenderer>(out var lr))
12706         {
12707             go.SetActive(false);
```

```
            _lines.Push(lr);
        }
        else if (go.TryGetComponent<Text>(out var tx))
        {
            go.SetActive(false);
            _texts.Push(tx);
            _liveTexts--;
        }
        else Destroy(go);
    }
    // ---------- 预设 ----------
    /// <summary>放大再淡出的一次性图。</summary>
    public void Burst(Sprite s, Vector2 pos, float size, float dur = 0.35f, int order = 9500,...
    {
        var col = c ?? Color.white;
        var sr = Take(s, pos, size * from, order, col, add);
        float rot0 = UnityEngine.Random.Range(0f, 360f) * (spin != 0 ? 1 : 0);
        Anim(sr.gameObject, dur, k =>
        {
            float sc = Mathf.Lerp(from, 1.08f, Tween.OutCubic(Mathf.Min(1, k * 1.8f)));
            sr.transform.localScale = Vector3.one * size * sc;
            if (spin != 0) sr.transform.localRotation = Quaternion.Euler(0, 0, rot0 + spin * k);
            sr.color = col.A(col.a * (k < 0.45f ? 1f : 1f - (k - 0.45f) / 0.55f));
        });
    }
    /// <summary>寨主剑气：金色月牙从 y0 推到 y1，宽度收进替天行道的半圆里。</summary>
    public void SwordWave(float y0, float y1, float speed, float width, Color c)
    {
        float travel = (y1 - y0) / speed;
        const float fade = 0.25f;
        float dur = travel + fade;
        Vector2 At(float t) => new V2(0, y0 + speed * Mathf.Min(t, travel)).ToV();
        float Alpha(float t) => t < 0.08f ? t / 0.08f : t > travel ? 1 - (t - travel) / fade ...
        float Wide(float y)
        {
            float dy = y - Layout.WallRowY;
            float r = Layout.HalfW;
            if (dy < 0f || dy >= r) return 0f;
            return Mathf.Min(width, 2f * Mathf.Sqrt(r * r - dy * dy));
        }
        var glow = Take(Art.Crescent, At(0), width, 9590, c.A(0.7f), true);
        var core = Take(Art.Crescent, At(0), width, 9591, Color.white, true);
        Anim(glow.gameObject, dur, k =>
        {
            float t = k * dur;
            float w = Wide(At(t).y);
            glow.transform.localPosition = At(t) + new Vector2(0, -0.15f);
            glow.transform.localScale = new Vector3(w * 1.05f, Mathf.Max(0.2f, w) * 1.2f, 1f);
            glow.color = c.A(0.75f * Alpha(t));
        });
```

```
12758         Anim(core.gameObject, dur, k =>
12759         {
12760             float t = k * dur;
12761             float w = Wide(At(t).y);
12762             core.transform.localPosition = At(t);
12763             core.transform.localScale = new Vector3(w, Mathf.Max(0.2f, w) * 1.05f, 1f);
12764             core.color = Color.white.A(0.95f * Alpha(t));
12765         });
12766         for (int i = 1; i <= 3; i++)
12767         {
12768             var ghost = Take(Art.Crescent, At(0), width, 9585, c.A(0), true);
12769             float lag = i * 0.06f;
12770             Anim(ghost.gameObject, dur, k =>
12771             {
12772                 float t = k * dur;
12773                 float tt = Mathf.Max(0, t - lag);
12774                 float w = Wide(At(tt).y);
12775                 ghost.transform.localPosition = At(tt);
12776                 ghost.transform.localScale = new Vector3(w, Mathf.Max(0.2f, w), 1f);
12777                 ghost.color = c.A(0.35f / i * Alpha(t) * (t > lag ? 1 : 0));
12778             });
12779         }
12780     }
12781     public void Ring(Vector2 pos, float radius, Color c, float dur = 0.4f, int order = 9400)
12782     {
12783         var sr = Take(Art.Ring, pos, 0.1f, order, c, true);
12784         Anim(sr.gameObject, dur, k =>
12785         {
12786             sr.transform.localScale = new Vector3(1f, 0.62f, 1f) * radius * 2f * Tween.OutCub...
12787             sr.color = c.A(c.a * (1 - k));
12788         });
12789     }
12790     /// <summary>地面光晕：柔光点压扁成椭圆。</summary>
12791     public void Glow(Vector2 pos, float size, Color c, float dur = 0.4f, int order = 140)
12792     {
12793         var sr = Take(Art.SoftDot, pos, size, order, c);
12794         Anim(sr.gameObject, dur, k =>
12795         {
12796             sr.transform.localScale = new Vector3(1f, 0.6f, 1f) * size * (0.7f + 0.5f * k);
12797             sr.color = c.A(c.a * (1 - k));
12798         });
12799     }
12800     /// <summary>天雷：底端落在 pos。</summary>
12801     public void Strike(Vector2 pos, float height, int order = 9600)
12802     {
12803         var sr = Take(Art.Img("fx_lightning"), pos + new Vector2(0, height * 0.5f), height, o...
12804         sr.flipX = UnityEngine.Random.value < 0.5f;
12805         Anim(sr.gameObject, 0.28f, k =>
12806         {
12807             float a = k < 0.15f ? 1f : Mathf.Abs(Mathf.Sin(k * 26f)) * (1 - k);
```

```
12808            sr.color = new Color(1, 1, 1, a);
12809            sr.transform.localScale = new Vector3(0.9f + 0.2f * k, 1f, 1f) * height;
12810         });
12811         var zap = Take(Art.Img("fx_zap"), pos + new Vector2(0, height * 0.12f), height * 0.5f...
12812         zap.flipX = UnityEngine.Random.value < 0.5f;
12813         Anim(zap.gameObject, 0.3f, k =>
12814         {
12815            zap.transform.localScale = new Vector3(1f, 0.75f, 1f) * height * Mathf.Lerp(0.35f...
12816            zap.color = Color.white.A(k < 0.2f ? 1f : 1f - (k - 0.2f) / 0.8f);
12817         });
12818         LightPool(pos, height * 1.1f, new Color(0.45f, 0.7f, 1f, 0.9f), 0.35f);
12819         Flash(pos + new Vector2(0, 0.4f), height * 0.9f, new Color(0.5f, 0.75f, 1f, 0.5f), 0....
12820         Decal(Art.Img("fx_scorch"), pos, height * 0.32f, new Color(0.12f, 0.12f, 0.2f, 0.55f)...
12821      }
12822      // ---------- 贴地：让特效“落”在地面上 ----------
12823      int _decals;
12824      /// <summary>地面印记（焦痕、裂缝、冰霜），压扁贴地，缓慢淡出。</summary>
12825      public void Decal(Sprite s, Vector2 pos, float size, Color c, float life = 2.5f, float fl...
12826      {
12827         if (_decals >= 40) return;
12828         _decals++;
12829         var sr = Take(s, pos, size, order, c.A(0), add);
12830         sr.flipX = UnityEngine.Random.value < 0.5f;
12831         float sx = size * UnityEngine.Random.Range(0.9f, 1.1f);
12832         sr.transform.localScale = new Vector3(sx, size * flat, 1f);
12833         Anim(sr.gameObject, life, k =>
12834         {
12835            float a = k < 0.05f ? k / 0.05f : k > 0.55f ? 1f - (k - 0.55f) / 0.45f : 1f;
12836            sr.color = c.A(c.a * a);
12837            if (k >= 1f) _decals--;
12838         });
12839      }
12840      /// <summary>贴地冲击波：压扁的不规则光环向外扩。</summary>
12841      public void GroundWave(Vector2 pos, float radius, Color c, float dur = 0.4f, int order = ...
12842      {
12843         var sr = Take(Art.Img("fx_wave"), pos, 0.1f, order, c, true);
12844         Anim(sr.gameObject, dur, k =>
12845         {
12846            float r = radius * 2f * Mathf.Lerp(0.2f, 1f, Tween.OutCubic(k));
12847            sr.transform.localScale = new Vector3(r, r * flat, 1f);
12848            sr.color = c.A(c.a * (1 - k) * (1 - k));
12849         });
12850      }
12851      /// <summary>地面受光：叠加的扁圆光斑，爆炸、落雷照亮地面。</summary>
12852      public void LightPool(Vector2 pos, float size, Color c, float dur = 0.4f, int order = 93)
12853      {
12854         var sr = Take(Art.SoftDot, pos, size, order, c, true);
12855         Anim(sr.gameObject, dur, k =>
12856         {
12857            sr.transform.localScale = new Vector3(1f, 0.55f, 1f) * size * (0.8f + 0.35f * Twe...
```

```
            sr.color = c.A(c.a * (k < 0.1f ? k / 0.1f : 1f - (k - 0.1f) / 0.9f));
        });
    }
    /// <summary>局部闪光：压在人物之上，连周围的兵一起照亮一下。</summary>
    public void Flash(Vector2 pos, float size, Color c, float dur = 0.16f, int order = 9450)
    {
        var sr = Take(Art.SoftDot, pos, size, order, c, true);
        Anim(sr.gameObject, dur, k =>
        {
            sr.transform.localScale = Vector3.one * size * (0.85f + 0.3f * k);
            sr.color = c.A(c.a * (1 - k) * (1 - k));
        });
    }
    /// <summary>烟团：普通混合，边升边胀边淡，给爆炸“分量”。</summary>
    public void Smoke(Vector2 pos, int n, float size, Color c, float dur = 1.1f, float spread...
    {
        var spr = Art.Img("fx_smoke");
        for (int i = 0; i < n; i++)
        {
            var off = new Vector2(UnityEngine.Random.Range(-spread, spread), UnityEngine.Rand...
            var p0 = pos + off;
            float sz = size * UnityEngine.Random.Range(0.7f, 1.15f);
            float rot = UnityEngine.Random.Range(0f, 360f);
            float spin = UnityEngine.Random.Range(-40f, 40f);
            float d = dur * UnityEngine.Random.Range(0.8f, 1.15f);
            var drift = new Vector2(off.x * 0.6f, rise * UnityEngine.Random.Range(0.7f, 1.2f));
            var sr = Take(spr, p0, sz * 0.4f, order + i, c.A(0));
            Anim(sr.gameObject, d, k =>
            {
                float e = Tween.OutCubic(k);
                sr.transform.localPosition = p0 + drift * e;
                sr.transform.localScale = Vector3.one * sz * Mathf.Lerp(0.45f, 1.25f, e);
                sr.transform.localRotation = Quaternion.Euler(0, 0, rot + spin * k);
                float a = k < 0.12f ? k / 0.12f : 1f - (k - 0.12f) / 0.88f;
                sr.color = c.A(c.a * a * a);
            }, true, i * 0.04f);
        }
    }
    /// <summary>贴地扬尘：横向铺开的低矮尘土。</summary>
    public void Dust(Vector2 pos, float width, Color c, float dur = 0.6f, int order = 9300)
    {
        var sr = Take(Art.Img("fx_dust"), pos, width * 0.5f, order, c);
        sr.flipX = UnityEngine.Random.value < 0.5f;
        Anim(sr.gameObject, dur, k =>
        {
            float e = Tween.OutCubic(k);
            float w = width * Mathf.Lerp(0.5f, 1.15f, e);
            sr.transform.localScale = new Vector3(w, w * 0.8f, 1f);
            sr.transform.localPosition = pos + new Vector2(0, w * 0.18f + 0.1f * e);
            sr.color = c.A(c.a * (1 - k));
```

```
12908         });
12909     }
12910     /// <summary>碎石：抛起、落地弹一下、停住后淡出。地面高度就是 pos.y 附近。</summary>
12911     public void Debris(Vector2 pos, int n, float size, float speed, float dur = 0.9f)
12912     {
12913         for (int i = 0; i < n; i++)
12914         {
12915             var spr = Art.Img("fx_rock" + UnityEngine.Random.Range(0, 6));
12916             float ang = UnityEngine.Random.Range(0f, Mathf.PI * 2);
12917             var vx = new Vector2(Mathf.Cos(ang), Mathf.Sin(ang) * 0.45f) * speed * UnityEngin...
12918             float vz = speed * UnityEngine.Random.Range(0.9f, 1.6f);
12919             float g = 14f;
12920             float t1 = 2f * vz / g;
12921             float sz = size * UnityEngine.Random.Range(0.55f, 1f);
12922             var ground = pos + new Vector2(0, UnityEngine.Random.Range(-0.25f, 0.15f));
12923             float spin = UnityEngine.Random.Range(-720f, 720f);
12924             var sr = Take(spr, ground, sz, 9350, Pal.Hex("8A7A68"));
12925             var sh = Take(Art.SoftDot, ground, sz, 125, new Color(0, 0, 0, 0.3f));
12926             float d = Mathf.Max(dur, t1 * 1.5f + 0.3f);
12927             Anim(sr.gameObject, d, k =>
12928             {
12929                 float t = k * d;
12930                 float z;
12931                 Vector2 xy;
12932                 if (t < t1) { z = vz * t - g * t * t * 0.5f; xy = vx * t; }
12933                 else
12934                 {
12935                     float t2 = t - t1, v2 = vz * 0.3f, tb = 2f * v2 / g;
12936                     z = t2 < tb ? v2 * t2 - g * t2 * t2 * 0.5f : 0f;
12937                     xy = vx * t1 + vx * 0.3f * Mathf.Min(t2, tb);
12938                 }
12939                 var gp = ground + xy;
12940                 sr.transform.localPosition = gp + new Vector2(0, z + sz * 0.3f);
12941                 sr.transform.localRotation = Quaternion.Euler(0, 0, spin * Mathf.Min(t, t1 * ...
12942                 sr.sortingOrder = Order(gp.y) + 30;
12943                 sh.transform.localPosition = gp;
12944                 sh.transform.localScale = new Vector3(sz * 0.9f, sz * 0.3f, 1f) / (1f + z * 0...
12945                 float fade = k > 0.75f ? 1f - (k - 0.75f) / 0.25f : 1f;
12946                 sr.color = Pal.Hex("8A7A68").A(fade);
12947                 sh.color = new Color(0, 0, 0, 0.3f * fade);
12948                 if (k >= 1f) Release(sh.gameObject);
12949             });
12950         }
12951     }
12952     /// <summary>从地面长出来的特效（冰晶、光柱）：底边钉在 pos，纵向拉起再收回。</summary>
12953     public void Rise(Sprite s, Vector2 pos, float size, float dur, Color c, int order, bool a...
12954     {
12955         var sr = Take(s, pos, size, order, c, add);
12956         sr.flipX = UnityEngine.Random.value < 0.5f;
12957         float aspect = s.bounds.size.y / Mathf.Max(s.bounds.size.x, s.bounds.size.y);
```

```
12958         Anim(sr.gameObject, dur, k =>
12959         {
12960             float up = k < 0.18f ? Tween.OutBack(k / 0.18f) : 1f;
12961             float fade = k > hold ? 1f - (k - hold) / (1f - hold) : 1f;
12962             float sy = size * up;
12963             sr.transform.localScale = new Vector3(size * (0.85f + 0.15f * up), sy, 1f);
12964             sr.transform.localPosition = pos + new Vector2(0, sy * aspect * 0.5f);
12965             sr.color = c.A(c.a * fade);
12966         });
12967     }
12968     /// <summary>落地残留物（插地的箭）：立在地面，停留后淡出。底边钉在 pos。</summary>
12969     public void Linger(Sprite s, Vector2 pos, float size, float life, bool flip = false)
12970     {
12971         if (_decals >= 40) return;
12972         _decals++;
12973         float aspect = s.bounds.size.y / Mathf.Max(s.bounds.size.x, s.bounds.size.y);
12974         var sr = Take(s, pos + new Vector2(0, size * aspect * 0.5f), size, Order(pos.y), Colo...
12975         sr.flipX = flip;
12976         Anim(sr.gameObject, life, k =>
12977         {
12978             float sq = k < 0.08f ? 1f + 0.15f * Mathf.Sin(k / 0.08f * Mathf.PI) : 1f;
12979             sr.transform.localScale = new Vector3(size / sq, size * sq, 1f);
12980             sr.color = Color.white.A(k > 0.6f ? 1f - (k - 0.6f) / 0.4f : 1f);
12981             if (k >= 1f) _decals--;
12982         });
12983     }
12984     /// <summary>全屏闪白/闪色，大招、首领落地用。</summary>
12985     public void ScreenFlash(Color c, float dur = 0.25f)
12986     {
12987         if (_cam == null) return;
12988         var sr = Take(Art.White, Vector2.zero, 1f, 9900, c, true);
12989         sr.transform.position = new Vector3(_cam.position.x, _cam.position.y, 0f);
12990         sr.transform.localScale = new Vector3(40f, 60f, 1f);
12991         Anim(sr.gameObject, dur, k => sr.color = c.A(c.a * (1 - k) * (1 - k)));
12992     }
12993     /// <summary>折线电弧。</summary>
12994     public void Arc(Vector2 a, Vector2 b, Color c, float width = 0.12f, float dur = 0.2f, int...
12995     {
12996         LineRenderer lr;
12997         if (_lines.Count > 0)
12998         {
12999             lr = _lines.Pop();
13000             lr.gameObject.SetActive(true);
13001         }
13002         else
13003         {
13004             var go = new GameObject("arc");
13005             go.transform.SetParent(transform, false);
13006             lr = go.AddComponent<LineRenderer>();
13007             lr.sharedMaterial = _lineMat;
```

```
13008            lr.useWorldSpace = true;
13009            lr.numCapVertices = 2;
13010            lr.positionCount = 8;
13011        }
13012        lr.sortingOrder = order;
13013        var d = b - a;
13014        var n = new Vector2(-d.y, d.x).normalized;
13015        float jag = Mathf.Min(0.35f, d.magnitude * 0.12f);
13016        for (int i = 0; i < 8; i++)
13017        {
13018            float t = i / 7f;
13019            float off = i == 0 || i == 7 ? 0 : UnityEngine.Random.Range(-jag, jag);
13020            lr.SetPosition(i, (Vector3)(a + d * t + n * off) + new Vector3(0, 0, -0.1f));
13021        }
13022        Anim(lr.gameObject, dur, k =>
13023        {
13024            float w = width * (1 - k * 0.6f);
13025            lr.startWidth = w;
13026            lr.endWidth = w * 0.7f;
13027            var cc = c.A(c.a * (1 - k));
13028            lr.startColor = cc;
13029            lr.endColor = Color.white.A(1 - k);
13030        });
13031    }
13032    /// <summary>从 a 飞到 b 的小图（金币、宝箱），带弧线。</summary>
13033    public void Fly(Sprite s, Vector2 a, Vector2 b, float size, float dur, float arc, Action ...
13034    {
13035        var sr = Take(s, a, size, order, Color.white);
13036        var side = new Vector2(UnityEngine.Random.Range(-0.8f, 0.8f), arc);
13037        Anim(sr.gameObject, dur, k =>
13038        {
13039            float e = Tween.InCubic(k) * 0.65f + k * 0.35f;
13040            var p = Vector2.Lerp(a, b, e) + side * Mathf.Sin(k * Mathf.PI) * (1 - k * 0.3f);
13041            sr.transform.localPosition = p;
13042            sr.transform.localScale = Vector3.one * size * (k < 0.15f ? Tween.OutBack(k / 0.1...
13043            if (k >= 1f) done?.Invoke();
13044        });
13045    }
13046    /// <summary>小碎片四散：命中火花、合成星星。</summary>
13047    public void Sparks(Sprite s, Vector2 pos, int n, float size, float speed, Color c, float ...
13048    {
13049        for (int i = 0; i < n; i++)
13050        {
13051            float ang = UnityEngine.Random.Range(0f, Mathf.PI * 2);
13052            var v = new Vector2(Mathf.Cos(ang), Mathf.Sin(ang) * 0.7f + 0.6f) * speed * Unity...
13053            float sz = size * UnityEngine.Random.Range(0.6f, 1f);
13054            var sr = Take(s, pos, sz, order, c, add);
13055            Anim(sr.gameObject, dur, k =>
13056            {
13057                float t = k * dur;
```

```
13058                sr.transform.localPosition = pos + v * t + new Vector2(0, -gravity * t * t * ...
13059                sr.transform.localScale = Vector3.one * sz * (1 - k * 0.7f);
13060                sr.transform.localRotation = Quaternion.Euler(0, 0, k * 240f);
13061                sr.color = c.A(c.a * (1 - k * k));
13062            });
13063        }
13064    }
13065    // ---------- 漂字 ----------
13066    public void Number(Vector2 world, string s, Color c, int size, float rise = 0.6f, float d...
13067    {
13068        Text t;
13069        if (_texts.Count > 0)
13070        {
13071            t = _texts.Pop();
13072            t.gameObject.SetActive(true);
13073            t.transform.SetAsLastSibling();
13074        }
13075        else
13076        {
13077            t = Ui.Label(_textLayer, "", 40, Color.white, Vector2.zero, new Vector2(500, 120)...
13078            Ui.Stroke(t, Pal.Ink, 3f);
13079            var rt0 = t.rectTransform;
13080            rt0.anchorMin = rt0.anchorMax = new Vector2(0.5f, 1f);
13081        }
13082        _liveTexts++;
13083        t.text = s;
13084        t.fontSize = size;
13085        t.color = c;
13086        var rt = t.rectTransform;
13087        var p0 = new Vector2(world.x * 100f, -Ui.FromTop(world.y));
13088        rt.anchoredPosition = p0;
13089        rt.localScale = Vector3.one;
13090        Anim(t.gameObject, dur, k =>
13091        {
13092            rt.anchoredPosition = p0 + new Vector2(0, rise * 100f * Tween.OutCubic(k));
13093            float sc = pop ? (k < 0.25f ? Mathf.Lerp(1.8f, 1f, k / 0.25f) : 1f) : (k < 0.15f ...
13094            rt.localScale = Vector3.one * sc;
13095            t.color = c.A(k < 0.6f ? 1f : 1f - (k - 0.6f) / 0.4f);
13096        });
13097    }
13098    /// <summary>跟随世界坐标的常驻小字（精英血条层数），调用方每帧 Place、用完自己 Destroy。</summary>
13099    public Text Pinned(int size, Color c)
13100    {
13101        var t = Ui.Label(_textLayer, "", size, c, Vector2.zero, new Vector2(160, 60), TextAnc...
13102        Ui.Stroke(t, Pal.Ink, 2.5f);
13103        var rt = t.rectTransform;
13104        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 1f);
13105        rt.pivot = new Vector2(0f, 0.5f);
13106        return t;
13107    }
```

```
13108       public static void Place(Text t, Vector2 world) =>
13109           t.rectTransform.anchoredPosition = new Vector2(world.x * 100f, -Ui.FromTop(world.y));
13110       // ---------- 震屏 ----------
13111       public void Shake(float amp, float dur = 0.25f)
13112       {
13113           if (amp < _shake * (_shakeT > 0 ? 1 : 0)) return;
13114           _shake = amp;
13115           _shakeT = dur;
13116       }
13117       void UpdateShake(float dt)
13118       {
13119           if (_cam == null) return;
13120           if (_shakeT <= 0)
13121           {
13122               _cam.localPosition = _camBase;
13123               return;
13124           }
13125           _shakeT -= dt;
13126           float a = _shake * Mathf.Clamp01(_shakeT / 0.25f);
13127           _cam.localPosition = _camBase + new Vector3(UnityEngine.Random.Range(-a, a), UnityEng...
13128       }
13129       public void SetCamBase(Vector3 p) => _camBase = p;
13130   }
13131 }
13132 // ===== FILE: Assets/Scripts/Sim/Bot.cs =====
13133 namespace Liangshan.Sim
13134 {
13135   /// <summary>自动玩家：数值验证和自动截图用。Skill 0 只会召唤，1 会合成，2 还会买强化、优先强化场上星级最高的好
13136   public sealed class Bot
13137   {
13138       readonly int _skill;
13139       float _think;
13140       public Bot(int skill) { _skill = skill; }
13141       public void Step(Battle b, float dt)
13142       {
13143           if (b.Phase == Phase.Choosing)
13144           {
13145               int pick = 0;
13146               if (_skill >= 2)
13147               {
13148                   int best = -1;
13149                   for (int i = 0; i < b.Choices.Length; i++)
13150                   {
13151                       int star = BestStar(b, b.Choices[i].Hero) + (b.Choices[i].Ult ? 10 : 0);
13152                       if (star > best) { best = star; pick = i; }
13153                   }
13154               }
13155               b.Pick(pick);
13156               return;
13157           }
```

```
            if (b.Phase == Phase.Reels)
            {
                b.ApplyReels(b.SpinReels());
                return;
            }
            if (b.Phase != Phase.Running) return;
            _think -= dt;
            if (_think > 0) return;
            _think = 0.35f;
            if (_skill >= 1 && TryMerge(b)) return;
            if (_skill >= 2 && b.FreeEnhances > 0 && b.Enhance()) return;
            if (_skill >= 2 && b.Coins >= b.EnhanceCost && b.EnhanceCost < b.SummonCost * 1.6f &&...
            if (b.CanSummon) { b.Summon(); return; }
            if (_skill >= 2 && !b.HasEmptySlot() && b.Coins >= b.EnhanceCost) b.Enhance();
        }
        static int BestStar(Battle b, HeroId h)
        {
            int best = 0;
            foreach (var s in b.Slots)
                if (s.Unit != null && s.Unit.Hero == h && s.Unit.Star > best) best = s.Unit.Star;
            return best;
        }
        static bool TryMerge(Battle b)
        {
            bool full = !b.HasEmptySlot();
            for (int star = 3; star >= 1; star--)
            {
                // 不满格时只合二星以上，留着一星占位输出。
                if (!full && star == 1 && b.Coins >= b.SummonCost) continue;
                for (int i = 0; i < b.Slots.Count; i++)
                {
                    var u = b.Slots[i].Unit;
                    if (u == null || u.Star != star) continue;
                    for (int j = i + 1; j < b.Slots.Count; j++)
                        if (b.CanMerge(i, j)) return b.Drop(i, j);
                }
            }
            return false;
        }
    }
}
// ===== FILE: Assets/Scripts/Sim/Core.cs =====
using System;
namespace Liangshan.Sim
{
    public struct V2
    {
        public float X, Y;
        public V2(float x, float y) { X = x; Y = y; }
        public static V2 operator +(V2 a, V2 b) => new V2(a.X + b.X, a.Y + b.Y);
```

```
13208       public static V2 operator -(V2 a, V2 b) => new V2(a.X - b.X, a.Y - b.Y);
13209       public static V2 operator *(V2 a, float k) => new V2(a.X * k, a.Y * k);
13210       public float Len => (float)Math.Sqrt(X * X + Y * Y);
13211       public float Len2 => X * X + Y * Y;
13212       public V2 Norm { get { float l = Len; return l > 1e-5f ? new V2(X / l, Y / l) : new V2(0,...
13213       public static float Dist(V2 a, V2 b) => (a - b).Len;
13214       public static V2 Lerp(V2 a, V2 b, float k) => new V2(a.X + (b.X - a.X) * k, a.Y + (b.Y - ...
13215   }
13216   /// <summary>xorshift，同一个种子一局完全复现。</summary>
13217   public sealed class Rng
13218   {
13219       uint _s;
13220       public Rng(int seed) { _s = (uint)seed * 2654435761u + 1u; if (_s == 0) _s = 1; }
13221       public uint Next()
13222       {
13223           _s ^= _s << 13;
13224           _s ^= _s >> 17;
13225           _s ^= _s << 5;
13226           return _s;
13227       }
13228       public float F() => (Next() & 0xFFFFFF) / 16777216f;
13229       public float Range(float a, float b) => a + (b - a) * F();
13230       public int Int(int n) => n <= 0 ? 0 : (int)(Next() % (uint)n);
13231       public bool Chance(float p) => F() < p;
13232   }
13233   public static class M
13234   {
13235       public static float Clamp(float v, float a, float b) => v < a ? a : v > b ? b : v;
13236       public static float Clamp01(float v) => Clamp(v, 0, 1);
13237       public static int Clamp(int v, int a, int b) => v < a ? a : v > b ? b : v;
13238   }
13239 }
13240 // ===== FILE: Assets/Scripts/Sim/MetaExtra.cs =====
13241 using System;
13242 namespace Liangshan.Sim
13243 {
13244   /// <summary>关卡三宝箱、天书、杏黄旗、装备品质、聚义宴、扫荡、寨主等级。</summary>
13245   public static partial class Meta
13246   {
13247       // ---------- 关卡三宝箱 ----------
13248       public static readonly string[] LevelChestCond = { "通关", "50%血量通关", "满血通关" };
13249       public static readonly float[] LevelChestWall = { 0f, 0.5f, 0.999f };
13250       public static bool LevelChestGot(SaveData s, int level, int k) => (s.LevelChest[level - 1...
13251       public static int LevelChestLeft(SaveData s, int level)
13252       {
13253           int n = 0;
13254           for (int k = 0; k < 3; k++) if (!LevelChestGot(s, level, k)) n++;
13255           return n;
13256       }
13257       /// <summary>第 n 关第 k 个宝箱。逢 5 的关卡满血箱换成玄女天书和玄铁。</summary>
```

```
13258     public static Rewards LevelChestReward(int n, int k)
13259     {
13260         var r = new Rewards();
13261         int ch = (n - 1) / Defs.LevelsPerChapter + 1;
13262         bool big = n % 5 == 0;
13263         switch (k)
13264         {
13265             case 0:
13266                 r.Add(Cur.Copper, 500 + 50 * n);
13267                 r.Add(Cur.Iron, 4 + n / 4);
13268                 r.Add(Cur.AnyShard, 2 + ch);
13269                 break;
13270             case 1:
13271                 r.Add(Cur.Page, 10 + n / 2);
13272                 r.Add(Cur.Silk, 100 + 10 * n);
13273                 r.Add(Cur.AnyShard, 3 + ch);
13274                 r.Add(Cur.Copper, 800 + 60 * n);
13275                 break;
13276             default:
13277                 if (big)
13278                 {
13279                     r.Add(Cur.Meteor, 2 + ch);
13280                     r.Add(Cur.Tome, ch <= 2 ? 2 : 3);
13281                 }
13282                 else r.Add(Cur.Page, 25 + n);
13283                 r.Add(Cur.Ingot, big ? 150 : 100);
13284                 r.Add(Cur.Silk, 200 + 20 * n);
13285                 r.Add(Cur.GearBox, 1);
13286                 r.Add(Cur.AnyShard, 4 + ch);
13287                 break;
13288         }
13289         return r;
13290     }
13291     // ---------- 扫荡 ----------
13292     /// <summary>三个宝箱都拿到的关卡可以扫荡：花体力直接拿一局胜利的常规奖励。</summary>
13293     public static bool CanSweep(SaveData s, int level) => level <= s.Cleared && s.LevelChest[...
13294     public static Rewards Sweep(SaveData s, LevelDef st, Rng rng)
13295     {
13296         if (!CanSweep(s, st.No) || !Spend(s, Cur.Stamina, BattleStamina)) return null;
13297         var r = BattleRewards(s, st, true, st.Waves, 1f, 0, rng);
13298         Give(s, r);
13299         s.Battles++;
13300         s.Wins++;
13301         Progress(s, 0);
13302         Progress(s, 1);
13303         return r;
13304     }
13305     // ---------- 天书（寨主技能） ----------
13306     public const int PageMax = 30;
13307     public static int PageCost(int lv) => 10 + 5 * lv;
```

```
13308       public static int PageCopper(int lv) => 500 + 200 * lv;
13309       public const float PageLord = 0.06f, PageDmg = 0.005f;
13310       public static bool CanPage(SaveData s) => s.PageLv < PageMax && s.Pages >= PageCost(s.Pag...
13311       public static bool LearnPage(SaveData s)
13312       {
13313           if (!CanPage(s)) return false;
13314           s.Pages -= PageCost(s.PageLv);
13315           s.Copper -= PageCopper(s.PageLv);
13316           s.PageLv++;
13317           return true;
13318       }
13319       public static readonly (string Name, string Desc, int Cost)[] TomeNodes =
13320       {
13321           ("剑气・二连", "替天行道连斩两道剑气", 2),
13322           ("剑气・震慑", "剑气命中的非首领敌人眩晕 0.8 秒", 3),
13323           ("剑气・斩将", "剑气对精英、首领伤害 +60%", 4),
13324           ("剑气・疾", "替天行道冷却 -20%", 5),
13325           ("剑气・三连", "替天行道连斩三道剑气", 6),
13326       };
13327       public static bool CanTome(SaveData s) => s.TomeNodes < TomeNodes.Length && s.Tomes >= To...
13328       public static bool LearnTome(SaveData s)
13329       {
13330           if (!CanTome(s)) return false;
13331           s.Tomes -= TomeNodes[s.TomeNodes].Cost;
13332           s.TomeNodes++;
13333           return true;
13334       }
13335       // ---------- 杏黄旗 ----------
13336       public const int FlagSlotsMax = 5, FlagMax = 10;
13337       public static int FlagSlots(int lv) => lv <= 0 ? 0 : Math.Min(FlagSlotsMax, 1 + lv / 2);
13338       public static int FlagMeteor(int lv) => lv + 1;
13339       public static int FlagCopper(int lv) => 2000 * (lv + 1);
13340       public static float FlagWall(int lv) => 0.02f * lv;
13341       public static float FlagStunSec(int lv) => lv <= 0 ? 0 : 2f + 0.2f * lv;
13342       public static int RerollCost(SaveData s)
13343       {
13344           int locked = 0;
13345           for (int i = 0; i < FlagSlotsMax; i++) if ((s.FlagLock & (1 << i)) != 0) locked++;
13346           return 200 + 200 * locked;
13347       }
13348       public const int FlagKinds = 6 + Defs.RoleCount;
13349       public static readonly string[] QualityName = { "普通", "精良", "史诗", "传说" };
13350       public static readonly float[] QualityBonus = { 0f, 0.15f, 0.3f, 0.5f };
13351       static readonly float[][] FlagVals =
13352       {
13353           new[] { 0.02f, 0.04f, 0.06f, 0.09f },
13354           new[] { 0.05f, 0.10f, 0.15f, 0.22f },
13355           new[] { 0.05f, 0.10f, 0.15f, 0.22f },
13356           new[] { 0.03f, 0.06f, 0.09f, 0.13f },
13357           new[] { 0.02f, 0.03f, 0.05f, 0.07f },
```

```
13358        new[] { 0.01f, 0.02f, 0.03f, 0.04f },
13359        new[] { 0.04f, 0.08f, 0.12f, 0.18f },
13360    };
13361    public static float FlagValue(int kind, int q) => FlagVals[Math.Min(kind, 6)][q];
13362    public static string FlagStatName(int kind) =>
13363        kind == 0 ? "全军伤害" : kind == 1 ? "对精英伤害" : kind == 2 ? "对首领伤害" : kind == 3 ? "寨墙生命" :...
13364        : Defs.RoleName((Role)(kind - 6)) + "伤害";
13365    public static bool AffixOf(SaveData s, int slot, out int kind, out int q)
13366    {
13367        int v = s.FlagAffix[slot] - 1;
13368        kind = v / 4;
13369        q = v % 4;
13370        return v >= 0;
13371    }
13372    public static string AffixText(int kind, int q) => $"{FlagStatName(kind)} +{FlagValue(kin...
13373    static int RollAffix(int lv, Rng rng)
13374    {
13375        int kind = rng.Int(FlagKinds);
13376        float w0 = 55 - 2 * lv, w1 = 30, w2 = 12 + lv, w3 = 3 + lv;
13377        float r = rng.F() * (w0 + w1 + w2 + w3);
13378        int q = r < w0 ? 0 : r < w0 + w1 ? 1 : r < w0 + w1 + w2 ? 2 : 3;
13379        return kind * 4 + q + 1;
13380    }
13381    public static bool CanFlagUp(SaveData s) => s.FlagLv < FlagMax && s.Meteor >= FlagMeteor(...
13382    /// <summary>打造 / 升阶：新开的词条位随机一条。</summary>
13383    public static bool FlagUp(SaveData s, Rng rng)
13384    {
13385        if (!CanFlagUp(s)) return false;
13386        s.Meteor -= FlagMeteor(s.FlagLv);
13387        s.Copper -= FlagCopper(s.FlagLv);
13388        s.FlagLv++;
13389        for (int i = 0; i < FlagSlots(s.FlagLv); i++)
13390            if (s.FlagAffix[i] == 0) s.FlagAffix[i] = RollAffix(s.FlagLv, rng);
13391        return true;
13392    }
13393    public static bool CanReroll(SaveData s)
13394    {
13395        if (s.FlagLv <= 0 || s.Silk < RerollCost(s)) return false;
13396        for (int i = 0; i < FlagSlots(s.FlagLv); i++) if ((s.FlagLock & (1 << i)) == 0) retur...
13397        return false;
13398    }
13399    /// <summary>洗练：没锁的词条全部重随。</summary>
13400    public static bool Reroll(SaveData s, Rng rng)
13401    {
13402        if (!CanReroll(s)) return false;
13403        s.Silk -= RerollCost(s);
13404        for (int i = 0; i < FlagSlots(s.FlagLv); i++)
13405            if ((s.FlagLock & (1 << i)) == 0) s.FlagAffix[i] = RollAffix(s.FlagLv, rng);
13406        return true;
13407    }
```

```
    public static void ToggleLock(SaveData s, int slot) => s.FlagLock ^= 1 << slot;
    // ---------- 装备箱 ----------
    public struct GearDrop
    {
        public int Slot, Quality, Iron;
        public bool Upgraded;
    }
    /// <summary>开一个装备箱：随机部位、品质（精良 65 / 史诗 28 / 传说 7）。比身上的好就换上，否则熔成精铁。</su
    public static GearDrop? OpenGearBox(SaveData s, Rng rng)
    {
        if (s.GearBoxes <= 0) return null;
        s.GearBoxes--;
        float r = rng.F() * 100;
        var d = new GearDrop { Slot = rng.Int(Equips.Length), Quality = r < 65 ? 1 : r < 93 ?...
        if (d.Quality > s.EquipQ[d.Slot])
        {
            s.EquipQ[d.Slot] = d.Quality;
            d.Upgraded = true;
        }
        else
        {
            d.Iron = 8 * (d.Quality + 1);
            s.Iron += d.Iron;
        }
        return d;
    }
    // ---------- 聚义宴（定时体力） ----------
    public static readonly int[] MealHours = { 6, 12, 18, 21 };
    public const int MealStamina = 15;
    public static int MealsReady(SaveData s, int hour)
    {
        int n = 0;
        for (int i = 0; i < MealHours.Length; i++) if (hour >= MealHours[i] && (s.Meals & (1 ...
        return n;
    }
    public static int EatMeals(SaveData s, int hour)
    {
        int n = 0;
        for (int i = 0; i < MealHours.Length; i++)
            if (hour >= MealHours[i] && (s.Meals & (1 << i)) == 0) { s.Meals |= 1 << i; n++; }
        s.Stamina += n * MealStamina;
        return n * MealStamina;
    }
    // ---------- 巡山锦缎 ----------
    public static float IdleSilkPerHour(SaveData s) => 20 + 2 * s.Cleared;
    // ---------- 寨主等级（只做展示） ----------
    public static int LordLevel(SaveData s, out float progress)
    {
        int exp = s.Wins * 20 + s.Cleared * 60 + s.Battles * 5;
        int lv = 1;
```

```
13458           while (exp >= 100 + 60 * (lv - 1)) { exp -= 100 + 60 * (lv - 1); lv++; }
13459           progress = exp / (float)(100 + 60 * (lv - 1));
13460           return lv;
13461       }
13462       // ---------- 汇总到战斗 ----------
13463       static void ApplyExtras(SaveData s, Loadout lo)
13464       {
13465           lo.LordPct *= 1f + PageLord * s.PageLv;
13466           lo.DmgMul = 1f + PageDmg * s.PageLv;
13467           if (s.TomeNodes >= 1) lo.LordHits = 2;
13468           if (s.TomeNodes >= 2) lo.LordStun = 0.8f;
13469           if (s.TomeNodes >= 3) lo.LordBig = 1.6f;
13470           if (s.TomeNodes >= 4) lo.LordCd *= 0.8f;
13471           if (s.TomeNodes >= 5) lo.LordHits = 3;
13472           lo.WallMul += FlagWall(s.FlagLv);
13473           lo.FlagStun = FlagStunSec(s.FlagLv);
13474           for (int i = 0; i < FlagSlots(s.FlagLv); i++)
13475           {
13476               if (!AffixOf(s, i, out int kind, out int q)) continue;
13477               float v = FlagValue(kind, q);
13478               switch (kind)
13479               {
13480                   case 0: lo.DmgMul += v; break;
13481                   case 1: lo.EliteDmg += v; break;
13482                   case 2: lo.BossDmg += v; break;
13483                   case 3: lo.WallMul += v; break;
13484                   case 4: lo.AllSpeed += v; break;
13485                   case 5: lo.Crit += v; break;
13486                   default: lo.RoleDmg[kind - 6] += v; break;
13487               }
13488           }
13489       }
13490   }
13491 }
13492 // ===== FILE: cloudfunctions/biehuang-api/index.js =====
13493 /**
13494  * 宋江别慌张后端（CloudBase 事件云函数 + HTTP 访问），照 songjiang-api。
13495  * gameKey：biehuang。
13496  *
13497  *   POST /login
13498  *   POST /save/pull
13499  *   POST /save/push
13500  *   GET|POST /health
13501  *
13502  * 环境变量：GAME_KEY=biehuang，BIEHUANG_JWT_SECRET 必填。
13503  * 微信真机再配 BIEHUANG_WX_APPID / BIEHUANG_WX_SECRET。
13504  * 集合：biehuang_playerData（userId 唯一索引）
13505  */
13506 const { handleLogin } = require('./lib/auth');
13507 const { handlePull, handlePush } = require('./lib/save');
```

```
const { respond, parseEvent, preflight } = require('./lib/http');
const ROUTES = {
  'GET /health': async () => ({ ok: true, ts: Date.now() }),
  'POST /health': async () => ({ ok: true, ts: Date.now() }),
  'POST /login': handleLogin,
  'POST /save/pull': handlePull,
  'POST /save/push': handlePush,
};
exports.main = async (event, context) => {
  try {
    if (event && event.httpMethod === 'OPTIONS') return preflight();
    const req = parseEvent(event);
    const key = `${req.method} ${req.path}`;
    const handler = ROUTES[key];
    if (!handler) {
      return respond(404, { ok: false, code: 'NOT_FOUND', error: `no route: ${key}` });
    }
    const result = await handler(req, context);
    if (result && typeof result === 'object' && 'statusCode' in result) return result;
    return respond(200, { ok: true, data: result });
  } catch (error) {
    const code = error && error.code ? error.code : 'INTERNAL';
    const status = error && error.status ? error.status : 500;
    const message = (error && error.message) || String(error);
    console.error('[biehuang-api] error:', code, message, error && error.stack);
    const out = { ok: false, code, error: message };
    if (error && error.data !== undefined) out.data = error.data;
    return respond(status, out);
  }
};
// ===== FILE: cloudfunctions/biehuang-api/lib/auth.js =====
const jwt = require('jsonwebtoken');
const { httpError } = require('./http');
const {
  getGameKey,
  gameKeyUpper,
  getJwtSecret: readJwtSecret,
  getTtlSec,
  getPlatformCredential,
} = require('./config');
const SUPPORTED_PLATFORMS = new Set(['wx', 'dy', 'anon']);
function getJwtSecret() {
  const secret = readJwtSecret();
  if (!secret) {
    throw httpError(500, 'NO_JWT_SECRET', `${gameKeyUpper()}_JWT_SECRET 未配置`);
  }
  return secret;
}
async function handleLogin(req) {
  const body = req.body || {};
```

```
  const platform = String(body.platform || '').toLowerCase();
  if (!SUPPORTED_PLATFORMS.has(platform)) {
    throw httpError(400, 'BAD_PLATFORM', `unsupported platform: ${platform}`);
  }
  let platformUid = '';
  if (platform === 'wx') {
    platformUid = (await wxCode2Openid(body.code)).openid;
  } else if (platform === 'dy') {
    platformUid = await ttCode2Openid(body.code);
  } else if (platform === 'anon') {
    const id = String(body.anonId || '').trim();
    if (!id) throw httpError(400, 'NO_ANON_ID', 'anonId 缺失');
    if (!/^[A-Za-z0-9_\-:.]{8,128}$/.test(id)) {
      throw httpError(400, 'BAD_ANON_ID', 'anonId 非法');
    }
    platformUid = id;
  }
  const userId = `${platform}:${platformUid}`;
  const ttlSec = getTtlSec();
  const gameKey = getGameKey();
  const now = Math.floor(Date.now() / 1000);
  const token = jwt.sign(
    { sub: userId, plt: platform, gk: gameKey, iat: now },
    getJwtSecret(),
    { expiresIn: ttlSec },
  );
  return {
    token,
    userId,
    platform,
    gameKey,
    expiresAt: (now + ttlSec) * 1000,
    ttlSec,
  };
}
function requireUser(req) {
  const authHeader = (req.headers && req.headers.authorization) || '';
  const match = /^Bearer\s+(.+)$/i.exec(authHeader);
  if (!match) throw httpError(401, 'NO_TOKEN', '缺少 Authorization: Bearer <token>');
  let payload;
  try {
    payload = jwt.verify(match[1].trim(), getJwtSecret());
  } catch (error) {
    throw httpError(401, 'BAD_TOKEN', error && error.message ? error.message : 'token 无效');
  }
  const userId = payload && payload.sub;
  if (!userId || typeof userId !== 'string' || !userId.includes(':')) {
    throw httpError(401, 'BAD_TOKEN', 'token sub 非法');
  }
  const currentGameKey = getGameKey();
```

```
  if (payload.gk && payload.gk !== currentGameKey) {
    throw httpError(401, 'BAD_TOKEN', `token gameKey=${payload.gk} 与当前 GAME_KEY=${currentGameKey}...
  }
  return { userId, platform: payload.plt || userId.split(':')[0] };
}
async function wxCode2Openid(code) {
  const appid = getPlatformCredential('wx', 'APPID');
  const secret = getPlatformCredential('wx', 'SECRET');
  if (!appid || !secret) throw httpError(500, 'NO_WX_CFG', `${gameKeyUpper()}_WX_APPID/${gameKeyU...
  if (!code) throw httpError(400, 'NO_CODE', 'wx code 缺失');
  const url = `https://api.weixin.qq.com/sns/jscode2session?appid=${encodeURIComponent(appid)}&se...
  const data = await httpGetJson(url);
  if (!data || !data.openid) {
    throw httpError(401, 'WX_LOGIN_FAIL', `wx code2session 失败: ${JSON.stringify(data || {})}`);
  }
  return { openid: data.openid, sessionKey: data.session_key || '' };
}
async function ttCode2Openid(code) {
  const appid = getPlatformCredential('tt', 'APPID');
  const secret = getPlatformCredential('tt', 'SECRET');
  if (!appid || !secret) throw httpError(500, 'NO_TT_CFG', `${gameKeyUpper()}_TT_APPID/${gameKeyU...
  if (!code) throw httpError(400, 'NO_CODE', 'dy code 缺失');
  const data = await httpPostJson('https://developer.toutiao.com/api/apps/v2/jscode2session', { a...
  if (!data || data.err_no !== 0 || !data.data || !data.data.openid) {
    throw httpError(401, 'TT_LOGIN_FAIL', `dy code2session 失败: ${JSON.stringify(data || {})}`);
  }
  return data.data.openid;
}
function httpGetJson(url) {
  return httpRequestJson(url, 'GET');
}
function httpPostJson(url, body) {
  return httpRequestJson(url, 'POST', body);
}
function httpRequestJson(url, method, body) {
  if (typeof fetch !== 'function') {
    return Promise.reject(httpError(500, 'NO_FETCH', '当前 Node 运行时不支持 fetch，请使用 Node 18+'));
  }
  const opts = { method, headers: { 'Content-Type': 'application/json' } };
  if (body !== undefined) opts.body = JSON.stringify(body);
  return fetch(url, opts).then(async (res) => {
    const text = await res.text();
    try {
      return JSON.parse(text);
    } catch (_) {
      return { _raw: text };
    }
  });
}
module.exports = {
```

```
  handleLogin,
  requireUser,
};
// ===== FILE: cloudfunctions/biehuang-api/lib/config.js =====
const DEFAULT_GAME_KEY = 'biehuang';
const DEFAULT_TTL_SEC = 7 * 24 * 3600;
const DEFAULT_MAX_BYTES = 256 * 1024;
function getGameKey() {
  const v = String(process.env.GAME_KEY || '').trim().toLowerCase();
  if (!v) return DEFAULT_GAME_KEY;
  if (!/^[a-z][a-z0-9_\-]{0,31}$/.test(v)) {
    throw new Error(`非法 GAME_KEY: ${v}`);
  }
  return v;
}
function gameKeyUpper() {
  return getGameKey().toUpperCase().replace(/[^A-Z0-9]/g, '_');
}
function readEnvPrefer(...keys) {
  for (const key of keys) {
    const value = process.env[key];
    if (value !== undefined && value !== null && String(value).length > 0) {
      return String(value);
    }
  }
  return '';
}
function getCollectionName(suffix) {
  const overrideKey = `${gameKeyUpper()}_${suffix.toUpperCase()}_COLLECTION`;
  const override = process.env[overrideKey];
  if (override) return String(override);
  return `${getGameKey()}_${suffix}`;
}
function getJwtSecret() {
  return readEnvPrefer(`${gameKeyUpper()}_JWT_SECRET`);
}
function getTtlSec() {
  const raw = readEnvPrefer(`${gameKeyUpper()}_TOKEN_TTL_SEC`);
  const v = Number(raw);
  return Number.isFinite(v) && v > 0 ? Math.floor(v) : DEFAULT_TTL_SEC;
}
function getMaxBytes() {
  const raw = readEnvPrefer(`${gameKeyUpper()}_SAVE_MAX_BYTES`);
  const v = Number(raw);
  return Number.isFinite(v) && v > 0 ? Math.floor(v) : DEFAULT_MAX_BYTES;
}
function getPlatformCredential(platform, field) {
  const upper = gameKeyUpper();
  const platformUpper = platform.toUpperCase();
  return readEnvPrefer(`${upper}_${platformUpper}_${field}`);
```

```
}
module.exports = {
  getGameKey,
  gameKeyUpper,
  readEnvPrefer,
  getCollectionName,
  getJwtSecret,
  getTtlSec,
  getMaxBytes,
  getPlatformCredential,
};
// ===== FILE: cloudfunctions/biehuang-api/lib/db.js =====
const tcb = require('@cloudbase/node-sdk');
const { getCollectionName } = require('./config');
let app = null;
function getApp() {
  if (app) return app;
  app = tcb.init({
    env: process.env.TCB_ENV || tcb.SYMBOL_CURRENT_ENV,
  });
  return app;
}
function getCollection() {
  return getApp().database().collection(getCollectionName('playerData'));
}
module.exports = {
  getApp,
  getCollection,
};
// ===== FILE: cloudfunctions/biehuang-api/lib/http.js =====
const { getGameKey } = require('./config');
const CORS_HEADERS = {
  'Access-Control-Allow-Origin': '*',
  'Access-Control-Allow-Methods': 'GET,POST,OPTIONS',
  'Access-Control-Allow-Headers': 'Content-Type,Authorization',
  'Access-Control-Max-Age': '86400',
};
function respond(statusCode, body, extraHeaders = {}) {
  return {
    statusCode,
    headers: {
      'Content-Type': 'application/json; charset=utf-8',
      ...CORS_HEADERS,
      ...extraHeaders,
    },
    body: typeof body === 'string' ? body : JSON.stringify(body),
    isBase64Encoded: false,
  };
}
function preflight() {
```

```
  return {
    statusCode: 204,
    headers: { ...CORS_HEADERS },
    body: '',
    isBase64Encoded: false,
  };
}
function parseEvent(event) {
  event = event || {};
  if (event.httpMethod) {
    const method = String(event.httpMethod).toUpperCase();
    let rawBody = event.body || '';
    if (event.isBase64Encoded && rawBody) {
      try {
        rawBody = Buffer.from(rawBody, 'base64').toString('utf8');
      } catch (_) {}
    }
    let body = {};
    if (rawBody) {
      try {
        body = JSON.parse(rawBody);
      } catch (_) {
        body = {};
      }
    }
    return {
      method,
      path: normalizePath(event.path || '/'),
      body,
      headers: lowercaseHeaders(event.headers || {}),
      query: event.queryStringParameters || {},
      raw: event,
    };
  }
  const action = String(event.action || '').replace(/^\/+/, '');
  return {
    method: 'POST',
    path: action ? `/${action}` : '/',
    body: event.body || {},
    headers: lowercaseHeaders(event.headers || {}),
    query: {},
    raw: event,
  };
}
function normalizePath(path) {
  if (!path) return '/';
  let p = String(path);
  if (!p.startsWith('/')) p = `/${p}`;
  const apiPrefix = `${getGameKey()}-api`;
  p = p.replace(new RegExp(`^/(?:${escapeRegExp(apiPrefix)})(?=/|$)`), '');
```

```
  if (p === '') p = '/';
  if (p.length > 1 && p.endsWith('/')) p = p.slice(0, -1);
  return p;
}
function lowercaseHeaders(headers) {
  const out = {};
  for (const key of Object.keys(headers || {})) {
    out[key.toLowerCase()] = headers[key];
  }
  return out;
}
function httpError(status, code, message) {
  const err = new Error(message || code);
  err.status = status;
  err.code = code;
  return err;
}
function escapeRegExp(text) {
  return String(text).replace(/[.*+?^${}()|[\]\\]/g, '\\$&');
}
module.exports = {
  respond,
  preflight,
  parseEvent,
  httpError,
};
// ===== FILE: cloudfunctions/biehuang-api/lib/save.js =====
const { httpError } = require('./http');
const { requireUser } = require('./auth');
const { getCollection } = require('./db');
const { getMaxBytes } = require('./config');
async function handlePull(req) {
  const { userId, platform } = requireUser(req);
  const col = getCollection();
  const res = await col.where({ userId }).limit(1).get();
  const doc = (res && Array.isArray(res.data) && res.data[0]) || null;
  if (!doc) {
    return {
      userId,
      platform,
      exists: false,
      schemaVersion: 0,
      updatedAt: 0,
      payload: {},
      payloadKeys: [],
    };
  }
  return {
    userId,
    platform,
```

```
    exists: true,
    schemaVersion: doc.schemaVersion || 0,
    updatedAt: doc.updatedAt || 0,
    payload: doc.payload || {},
    payloadKeys: Array.isArray(doc.payloadKeys) ? doc.payloadKeys : Object.keys(doc.payload || {}),
    clientFingerprint: doc.clientFingerprint || '',
  };
}
async function handlePush(req) {
  const { userId, platform } = requireUser(req);
  const body = req.body || {};
  const schemaVersion = Number(body.schemaVersion);
  const updatedAt = Number(body.updatedAt);
  const baseRemoteUpdatedAt = Number(body.baseRemoteUpdatedAt || 0);
  const payload = body.payload;
  const clientFingerprint = String(body.clientFingerprint || '').slice(0, 200);
  if (!Number.isFinite(schemaVersion) || schemaVersion <= 0) {
    throw httpError(400, 'BAD_SCHEMA', 'schemaVersion 非法');
  }
  if (!Number.isFinite(updatedAt) || updatedAt <= 0) {
    throw httpError(400, 'BAD_UPDATED_AT', 'updatedAt 非法');
  }
  if (!Number.isFinite(baseRemoteUpdatedAt) || baseRemoteUpdatedAt < 0) {
    throw httpError(400, 'BAD_BASE_REMOTE_UPDATED_AT', 'baseRemoteUpdatedAt 非法');
  }
  if (!payload || typeof payload !== 'object' || Array.isArray(payload)) {
    throw httpError(400, 'BAD_PAYLOAD', 'payload 必须是 object');
  }
  const size = Buffer.byteLength(JSON.stringify(payload), 'utf8');
  const maxBytes = getMaxBytes();
  if (size > maxBytes) {
    throw httpError(413, 'PAYLOAD_TOO_LARGE', `payload 超限: ${size}B > ${maxBytes}B`);
  }
  const payloadKeys = [];
  for (const k of Object.keys(payload)) {
    if (typeof payload[k] !== 'string') {
      throw httpError(400, 'BAD_PAYLOAD_VALUE', `payload[${k}] 必须是字符串`);
    }
    payloadKeys.push(k);
  }
  const col = getCollection();
  const existingRes = await col.where({ userId }).limit(1).get();
  const existing = (existingRes && Array.isArray(existingRes.data) && existingRes.data[0]) || null;
  if (existing) {
    const prevUpdatedAt = Number(existing.updatedAt) || 0;
    if (updatedAt < prevUpdatedAt || baseRemoteUpdatedAt < prevUpdatedAt) {
      throw Object.assign(
        httpError(409, 'STALE_UPDATE', `服务端已有更新版本 remote=${prevUpdatedAt} > local=${updatedAt}, b...
        {
          data: {
```

```
          remote: {
            schemaVersion: existing.schemaVersion || 0,
            updatedAt: prevUpdatedAt,
            payload: existing.payload || {},
            payloadKeys: Array.isArray(existing.payloadKeys)
              ? existing.payloadKeys
              : Object.keys(existing.payload || {}),
          },
        },
      },
    );
  }
}
const mergedPayload = { ...(existing && existing.payload ? existing.payload : {}) };
for (const k of Object.keys(payload)) mergedPayload[k] = payload[k];
const mergedPayloadKeys = Object.keys(mergedPayload);
const mergedSize = Buffer.byteLength(JSON.stringify(mergedPayload), 'utf8');
if (mergedSize > maxBytes) {
  throw httpError(413, 'PAYLOAD_TOO_LARGE', `payload 超限: ${mergedSize}B > ${maxBytes}B`);
}
const now = Date.now();
const docData = {
  userId,
  platform,
  schemaVersion,
  updatedAt,
  baseRemoteUpdatedAt,
  clientFingerprint,
  payload: mergedPayload,
  payloadKeys: mergedPayloadKeys,
  lastWriteAt: now,
};
if (existing && existing._id) {
  await col.doc(existing._id).update(docData);
  return { userId, updatedAt, savedAt: now, mode: 'update', sizeBytes: mergedSize };
}
const addRes = await col.add(docData);
return {
  userId,
  updatedAt,
  savedAt: now,
  mode: 'insert',
  sizeBytes: size,
  _id: addRes && (addRes.id || addRes._id),
};
}
module.exports = {
  handlePull,
  handlePush,
};
```